

Traceability Matrix

REMEMBER GORDON! – Rulebook $\rightleftharpoons$ Implementation Mapping

Generated from `docs/traceability.toml`

Overview

Implemented	Descriptive	Implicit	Out-of-scope
85	32	4	3

Total mappings: 124 · Total impl sites: 252

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 1/3 implemented

§1 – Introduction

§1 – Introduction

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Introduction

§1.1 – General Comments

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manual page 1

General Comments

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is a simulation of the final battle in Great Britain’s two-year campaign to reassert her presence in the Sudan (1896–1898). Fought September 2nd, 1898, Omdurman finally broke the back of the fanatical Dervish rebellion and gained Britain a million square miles of desolate territory and two million impoverished subjects. With two players, one assumes the role of Herbert Kitchener, Sirdar (CIC) of the Anglo-Egyptian army; the other player becomes the Khalifa, Abdullah the Taiasha, absolute ruler of the Dervishes. The game is also suited for multi-player participation, with each player assuming command of one or more Dervish tribes or Anglo-Egyptian brigades.

While “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is not, strictly speaking, a beginner’s game, the mechanics of play should be familiar to players of modest experience. It is suggested that the bonus game, FALL OF KHARTOUM, and the historical scenario be played first to familiarize players with the game system prior

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/unit_profiles.rs:194 GH:omdurman-rules/src/unit_profiles.rs:194	command_owns_unit	<pre>192 /// treat *nobody-claims-it* units as the faction's communal pool, so `false` 193 /// here means "any member of the faction may act on this unit". 194 pub fn command_owns_unit(scope: &omdurman_types::CommandScope, identity: &UnitIdentity) -> bool { 195 match scope { 196 omdurman_types::CommandScope::Army => false,</pre>

Covered by tests:

- [omdurman-rules::src::unit_profiles::command_owns_unit_follows_tribes_and_brigades](#)
- [omdurman-net::src::lib::start_game_commands_round_trip](#)
- [omdurman-net::src::lib::legacy_start_game_without_commands_still_loads](#)
- [omdurman-net::src::lib::brigade_scope_display_and_membership](#)
- [omdurman-app::src::tests::replayed_start_game_stages_commands_and_resets_ready](#)

§1.2 – Game Scale

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Game Scale

Each hexagon of the mapsheet represents approximately 400–440 yards of real terrain and each day turn is the equivalent of two hours of real time. Each counter of infantry and cavalry represents between 400 and 700 men, and each of the gunboats present at the battle has its own counter. The upper echelon of command is represented by individual leader counters for the Anglo-Egyptian force; and leaders plus their retinues for the Dervish army.

§2 – Game Components

§2 – Game Components

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Game Components

§2.1 – The Game Maps

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The Game Maps

The Omdurman battle map represents approximately 100 square miles of real territory and includes the area north of Omdurman in which the historical battle took place as well as the dominant terrain features that influenced the course of the battle. Note that the mapsheet also contains the Turn Record Track, Turn Sequence, and Terrain Effects Chart at the top; and the Combat Tables and Howitzer Fire Scattergram in the lower right corner. The large letters “A”, “D”, “Y”, etc. are set-up hexes for the historical scenario only (9.2) and should be ignored in the campaign game. Similarly, the hexsides of the Zariba exist only in the historical scenario and should be considered clear terrain in the campaign game. Note, however, that the Anglo-Egyptian player may “construct” the Zariba in the campaign game if desired (see 5.3). All full hexes of the Omdurman game map are playable, including the seven hexes of the Howitzer Fire Scattergram.

The mini-map for the bonus game, FALL OF KHARTOUM, represents that city as it appeared in 1885. The portion (see manual for full text)

See also: §5.3, §9.2

§2.2 – Play Aids

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Play Aids

Certain charts and tables are needed to play the game. The Terrain Effects Chart lists all terrain found on the mapsheet and the effect of each type on movement and combat. The Combat Tables describe the range effects on various weapon types and includes the Combat Results Table. Also note the Line of Sight Table on the back of this rulebook. It tells players when certain terrain types block line of sight, thus preventing direct fire attacks on enemy units. Players should become familiar with these various charts and tables prior to the beginning of play.

(see manual for full text)

§2.3 – The Units

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The Units

File	Symbol	Code Snippet
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omdurman-types/src/lib.rs:882 GH:omdurman-types/src/lib.rs:882	UnitKind	<pre> 880 /// `Some(UnitKind::Marker)` or `None`. 881 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 882 pub enum UnitKind { 883 /// Foot infantry (\$2.3): fire / melee / movement. 884 Infantry { </pre>
omdurman-rules/src/lib.rs:798 GH:omdurman-rules/src/lib.rs:798	UnitProfile	<pre> 796 /// print no melee value). 797 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 798 pub struct UnitProfile { 799 pub kind: UnitKind, 800 pub identity: UnitIdentity, </pre>
omdurman-rules/src/lib.rs:16 GH:omdurman-rules/src/lib.rs:16	BrigadeId	<pre> 14 15 use omdurman_types::{ 16 BrigadeId, BrigadeNationality, DayNight, DervishTribe, Faction, HexCoord, HexsideRef, Player, 17 SetupLetter, UnitKind, 18 }; </pre>
omdurman-types/src/lib.rs:679 GH:omdurman-types/src/lib.rs:679	SpriteAnnotation	<pre> 677 /// as an optional overlay over the compiled `sprite_data` fallback. 678 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq)] 679 pub struct SpriteAnnotation { 680 pub color: SpriteColor, 681 pub faction: Option<Faction>, </pre>

Covered by tests:

- [omdurman-rules::src::unit_profiles::british_army_row_zero_specials_classify_by_counter](#)
- [omdurman-rules::src::unit_profiles::egyptian_army_row_zero_specials_classify_by_counter](#)
- [omdurman-rules::src::unit_profiles::tribe_stats_come_from_annotation](#)
- [omdurman-rules::src::unit_profiles::section_owner_dervish_sections](#)
- [omdurman-rules::src::unit_profiles::section_owner_green_sections_are_dervish](#)

§2.4 – Game Parts Inventory

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Game Parts Inventory

Your complete copy of “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN includes:

- One 22 × 28 Battle of Omdurman mapsheet
- One 8½ × 11 bonus game: FALL OF KHARTOUM mapsheet
- One Rules Booklet
- One die-cut Unit Counter Sheet
- One Campaign Game Order of Appearance Card
- One ten-sided die
- One game box

§2.31 – Dervish weapon types

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Dervish artillery, gunboats, and forts fire on the “artillery” line of the Dervish Range Effects Table; Jehadia and Danagla units fire on the “rifles” line as does the Isa Zachneih unit. All other Dervish units (including leaders) are armed with spears and swords.

File	Symbol	Code Snippet
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omdurman-rules/src/lib.rs:476 GH:omdurman-rules/src/lib.rs:476	WeaponClass	<pre> 474 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, 475 PartialOrd, Ord, Hash, Debug, strum::Display, 476 pub enum WeaponClass { 477 /// Dervish spears and swords -- no ranged fire at 478 all. 479 Melee, </pre>
omdurman-rules/src/unit_profiles.rs:578 GH:omdurman-rules/src/unit_profiles.rs:578	dervish_tribe	<pre> 576 /// Resolve a Dervish tribal foot counter (§2.31): 577 /// Isa Zachneih fire on the rifles line; every other 578 tribe is spear-armed. 579 pub fn dervish_tribe(tribe: DervishTribe) -> 580 Option<Classification> { 581 // §2.31: "Jehadia and Danagla units fire on the 582 'rifles' line as does the 583 // Isa Zachneih unit. All other Dervish units 584 (including leaders) are armed </pre>
omdurman-rules/src/unit_profiles.rs:344 GH:omdurman-rules/src/unit_profiles.rs:344	khalifa_abdullah	<pre> 342 /// battle (§9.322). All three are interchangeable, 343 so they share the 344 `DervishArtillery` identity. 345 pub fn khalifa_abdullah(col: u32, row: u32) -> 346 Option<Classification> { 347 let artillery = { 348 Some(Classification { </pre>

Covered by tests: [omdurman-rules::src::unit_profiles::dervish_weapon_class_follows_the_rifles_line](#)

§2.32 – Anglo-Egyptian weapon types

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All Anglo-Egyptian units (except gunboats, Maxims, artillery, and leaders) are armed with rifles. Maxims fire on the “Maxims” line of the Anglo-Egyptian Range Effects Table, and artillery and old gunboats fire on the “Artillery” line. New type (named) gunboats may fire on the “Howitzer”, “Artillery”, and “Maxims” lines of the Range Effects Table. (See 6.52 for the fire capabilities of the “Friendlies”).

****Sample Dervish Units**** (printed on counters): combat unit (Combat / Melee / Movement values, plus Tribe identifier); Leader (e.g. OSMAN DIGNA); Camel unit (e.g. Danagla, 4-6-12); Fort.

****Sample Anglo-Egyptian Units**** (printed on counters): Cavalry (e.g. 21 Lancers); Artillery (e.g. 32 Battery); Old Gunboat (e.g. LORD KITCHENER, 0-0-15); New Gunboat — named (e.g. Sultan, with artillery and howitzer factor, plus movement downstream / movement upstream values); Maxim Guns (fire twice per turn); Infantry (Fire Combat Factor / Melee / Movement, plus Battalion ID and Brigade ID — e.g. “2B” = 2nd British Brigade, “3E” = 3rd Egyptian Brigade).

(see manual for full text)

See also: §6.52

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:428 GH:omdurman-rules/src/lib.rs:428	GunboatId	<pre> 426 /// fire; "old" gunboats do not (rulebook §2.32). 427 #[derive(Serialize, Deserialize, Clone, Copy, 428 PartialEq, Eq, Hash, Debug, strum::Display)] 429 pub enum GunboatId { 430 /// One of the five new-type named gunboats with 431 howitzer capability. 432 Named(NamedGunboat), </pre>
omdurman-rules/src/lib.rs:447 GH:omdurman-rules/src/lib.rs:447	NamedGunboat	<pre> 445 /// The five named gunboats with howitzer capability 446 (rulebook §6.64, §2.32). 447 #[derive(Serialize, Deserialize, Clone, Copy, 448 PartialEq, Eq, Hash, Debug, strum::Display)] 449 pub enum NamedGunboat { 450 Sultan, 451 Melik, </pre>

omdurman-rules/src/lib.rs:460 GH:omdurman-rules/src/lib.rs:460	OldGunboat	<pre>458 /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 459 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 460 pub enum OldGunboat { 461 LordKitchener, 462 Tamai,</pre>
omdurman-rules/src/lib.rs:460 GH:omdurman-rules/src/lib.rs:460	GunboatId::Old	<pre>458 /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 459 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 460 pub enum OldGunboat { 461 LordKitchener, 462 Tamai,</pre>
omdurman-rules/src/lib.rs:434 GH:omdurman-rules/src/lib.rs:434	GunboatId::Dervish-Gunboat	<pre>432 Old(OldGunboat), 433 /// A Dervish gunboat (\$9.111, \$10.14). 434 DervishGunboat(u8), 435 } 436 </pre>

Covered

by

tests:

omdurman-rules::src::effects::tests::old_gunboat_lacks_howitzer

omdurman-rules::src::effects::tests::old_gunboat_rejected_from_howitzer_subphase

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| §3 – Getting Started

§3 – Getting Started

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Getting Started

Spread out the mapsheet on a table. It should lie flat if you backfold it against the scored lines. The Dervish player should sit next to the west edge of the map and the Anglo-Egyptian player opposite him on the east edge. Read through the rules once, looking over the various charts as they are referred to in the various sections. Next, select a scenario and punch out only those unit counters needed to play. Later on, the rest of the unit counters should be punched out, sorted and stored by unit type.

(see manual for full text)

■ 1/1 implemented

§4 – Turn Sequence

§4 – Turn Sequence

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Turn Sequence

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is played in “Game Turns”, each of which has two “Player Turns”. The player moving first will vary according to the scenario being played. In the Campaign Game, for example, the Anglo-Egyptian player moves first.

A) Anglo-Egyptian Player Turn:

1. Anglo-Egyptian Movement Phase
2. Fire Combat Phase a. Dervish Defensive Fire b. Anglo-Egyptian Offensive Fire
 1. Direct Fire Subphase
 2. Maxim Second Fire and Howitzer Fire Subphase

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:252 GH:omdurman-rules/src/lib.rs:252	GameTurnIndex	<pre>250 /// One-based Game Turn index (1, 2, ... up to the 251 scenario length) (rulebook §4). 252 #[derive(Serialize, Deserialize, Clone, Copy, 253 PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 254 pub struct GameTurnIndex(u8); 255 256 impl GameTurnIndex {</pre>
omdurman-rules/src/lib.rs:274 GH:omdurman-rules/src/lib.rs:274	Phase	<pre>272 /// etc. 273 #[derive(Serialize, Deserialize, Clone, Copy, 274 PartialEq, Eq, Hash, Debug, Default)] 275 pub enum Phase { 276 /// Pre-game deployment (§9.2/§9.3/§10): fixed 277 units are placed, each side 278 /// deploys its order of battle within its legal 279 zone, and river</pre>
omdurman-rules/src/effects/state.rs:5 GH:omdurman-rules/src/effects/state.rs:5	GameState	<pre>3 /// All mutable state of a game in progress (rulebook 4 §4). 5 #[derive(Serialize, Deserialize, Clone, Debug)] 6 pub struct GameState { 7 pub scenario: Scenario, 8 pub current_turn: GameTurnIndex,</pre>
omdurman-rules/src/effects/state.rs:156 GH:omdurman-rules/src/effects/state.rs:156	GameState::new	<pre>154 impl GameState { 155 /// Create a fresh game state for a given scenario 156 (rulebook §4). 157 pub fn new(scenario: Scenario) -> Self { 158 let first = scenario_turn(scenario, 159 GameTurnIndex::new(1)); 160 // First player to *move* per scenario: 161 Campaign -- Anglo-Egyptian moves</pre>
omdurman-rules/src/effects/effect.rs:17 GH:omdurman-rules/src/effects/effect.rs:17	AdvancePhase	<pre>15 /// At end-of-turn, disrupted units recover and 16 per-turn tracking is 17 /// cleared. 18 AdvancePhase, 19 // -- Movement 20 -----</pre>
omdurman-rules/src/effects/dispatch.rs:98 GH:omdurman-	advance_phase	<pre>96 97 /// Advance the game state to the next phase (rulebook 98 §4).</pre>

rules/src/effects/ dispatch.rs:98		<pre> 98 pub fn advance_phase(state: &mut GameState) -> Result<(), RuleError> { 99 let old_phase = state.phase; 100 #[cfg(not(feature = "kani"))]</pre>
omdurman- rules/src/effects/ dispatch.rs:204 GH:omdurman- rules/src/effects/ dispatch.rs:204	end_player_turn	<pre> 202 203 /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook §4). 204 pub fn end_player_turn(state: &mut GameState) -> Result<(), RuleError> { 205 #[cfg(not(feature = "kani"))] 206 debug!(</pre>
omdurman- rules/src/ lib.rs:63 GH:omdurman- rules/src/lib.rs:63	GameTurnIndex::value	<pre> 61 pub const ALL: &'static [Self] = &{\$(Self::: \$variant,)+}; 62 63 pub fn value(self) -> u16 { 64 match self { 65 \$(Self::\$variant => \$value,)+</pre>
omdurman- rules/src/effects/ state.rs:148 GH:omdurman- rules/src/effects/ state.rs:148	PendingMelee	<pre> 146 /// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5). 147 #[derive(Serialize, Deserialize, Clone, Debug)] 148 pub struct PendingMelee { 149 pub attack: MeleeAttack, 150 pub attacker_roll: DieRoll,</pre>

Covered by tests:

- omdurman-rules::src::effects::tests::both_ready_auto_advances_out_of_setup
- omdurman-rules::src::effects::tests::fire_combat_wrong_phase_rejected
- omdurman-rules::src::effects::tests::new_game_starts_in_setup
- omdurman-rules::src::turn_track::scenario_turn_dispatches_correctly
- omdurman-rules::src::effects::tests::turn_advances_through_phases
- omdurman-rules::src::unit_profiles::game_turn_marker_cell_returns_none

18/23 implemented

§5 – Movement Phase

§5 – Movement Phase (general)

descriptive

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Movement Phase

§5.1 – General Rules

descriptive

manual page unknown

General Rules

§5.2 – Movement Restrictions

descriptive

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Movement Restrictions

§5.3 – Constructing the Zariba

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Constructing the Zariba

The Zariba trench and thorn hedge hexsides are built and in place in the historical scenario only. These hexsides are considered clear terrain in the campaign game. The Anglo-Egyptian player may, however, find it useful to construct this defensive position during the campaign game. The Zariba hexsides may only be built in their position as displayed on the mapsheet. Construction procedure is as follows: any Anglo-Egyptian infantry unit that begins and ends the Anglo-Egyptian player turn adjacent to (and on the Nile side of) Zariba hexsides has constructed all Zariba hexsides to which he is adjacent. The constructing unit may neither fire offensively nor melee attack during the turn of construction. Use a blank counter to denote units constructing Zariba hexsides. See 9.23 for defensive benefits and movement restrictions of Zariba hexsides.

(see manual for full text)

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:832 GH:omdurman-rules/src/lib.rs:832	constructing_zariba	<pre>830 /// Set while the unit is building Zariba hexsides -- -- neither offensive 831 /// fire nor melee allowed that turn (§5.3). 832 pub constructing_zariba: bool, 833 /// Set when the Royal Engineers are committed to a demolition this turn 834 /// (§6.53) -- neither offensive fire nor melee allowed that turn.</pre>
omdurman-rules/src/effects/effect.rs:169 GH:omdurman-	ConstructZariba	<pre>167 168 /// Begin constructing a Zariba hexside (rulebook §5.3). 169 ConstructZariba {</pre>

rules/src/effects/ effect.rs:169		<pre> 170 unit_ids: Vec<UnitId>, 171 hexside: HexsideRef, </pre>
omdurman-rules/src/effects/dispatch.rs:40 GH:omdurman-rules/src/effects/dispatch.rs:40	apply_construct_zariba	<pre> 38 GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id), 39 GameEffect::ConstructZariba { unit_ids, hexside } => { 40 apply_construct_zariba(state, unit_ids, *hexside) 41 } 42 GameEffect::Demolition { unit_id, target } => apply_demolition(state, *unit_id, *target), </pre>

Covered by tests: [omdurman-rules::src::effects::tests::construct_zariba_marks_builders_and_records_hexside](#)
[omdurman-rules::src::effects::tests::world_lens_composes_breaches_and_constructed_zariba](#)

§5.4 – Zones of Control

[descriptive](#)

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Zones of Control

§5.5 – Stacking

[descriptive](#)

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Stacking

§5.11 – Movement allowances printed on units

[implemented](#)

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The movement allowances of the various unit types are printed directly on the units (see 2.3). A unit may move up to this printed movement allowance, paying varying costs for different terrain types (see the Terrain Effects Chart).

See also: §2.3

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:131 GH:omdurman-rules/src/lib.rs:131	MovementAllowance	<pre> 129 /// is a named variant. 130 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 131 pub enum MovementAllowance { 132 /// Immobile (forts, wrecked gunboats). 133 Immobile = 0, </pre>
omdurman-rules/src/lib.rs:809 GH:omdurman-rules/src/lib.rs:809	UnitMovement	<pre> 807 /// Movement allowance -- uniform for land units, split for gunboats (rulebook §5.11, §5.24, §5.25). 808 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 809 pub enum UnitMovement { 810 Land(MovementAllowance), 811 Gunboat(GunboatMovement), </pre>
omdurman-types/src/lib.rs:302 GH:omdurman-types/src/lib.rs:302	HexDirection	<pre> 300 /// (`+q`, `+q+r`, `+r`, `-q`, `-q-r`, `-r` for pointy- top hexes) (rulebook §5.11, §5.24). 301 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, Default)] 302 pub enum HexDirection { 303 #[default] 304 East = 0, </pre>

omdurman-rules/src/lib.rs:173 GH:omdurman-rules/src/lib.rs:173	MovementPoints	<pre> 171 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, 172 PartialOrd, Ord, Hash, Debug, Default, 173 pub struct MovementPoints(i16); 174 175 impl MovementPoints { </pre>
omdurman-rules/src/terrain_chart.rs:21 GH:omdurman-rules/src/terrain_chart.rs:21	terrain_effects_chart	<pre> 19 /// 20 /// Source: printed Terrain Effects Chart on the mapsheet. 21 pub fn terrain_effects_chart(terrain: Terrain) -> TerrainEntry { 22 match terrain { 23 Terrain::Clear { .. } => TerrainEntry { </pre>
omdurman-rules/src/terrain_chart.rs:77 GH:omdurman-rules/src/terrain_chart.rs:77	movement_cost	<pre> 75 /// Convenience: get the movement cost for a terrain type (rulebook §5.11, Terrain Effects Chart). 76 /// Returns `None` for impassable terrain (Nile). 77 pub fn movement_cost(terrain: Terrain) -> Option<MovementAllowance> { 78 terrain_effects_chart(terrain).movement_cost 79 } </pre>
omdurman-rules/src/terrain_chart.rs:85 GH:omdurman-rules/src/terrain_chart.rs:85	movement_cost_with_road	<pre> 83 /// underlying terrain; without a road it's the terrain's own cost. The road is 84 /// a movement overlay only -- combat/LOS still use the underlying terrain. 85 pub fn movement_cost_with_road(terrain: Terrain, road: bool) -> Option<MovementAllowance> { 86 if road { 87 Some(MovementAllowance::One) </pre>
omdurman-types/src/lib.rs:498 GH:omdurman-types/src/lib.rs:498	Terrain::has_road	<pre> 496 497 /// Whether this hex has any road touching it. 498 pub fn has_road(self) -> bool { 499 !matches!(self.road(), Road::None) 500 } </pre>
omdurman-types/src/lib.rs:435 GH:omdurman-types/src/lib.rs:435	Terrain::passable_by_land	<pre> 433 434 /// Whether this terrain may be entered by land units (rulebook §5.11). 435 pub fn passable_by_land(self) -> bool { 436 !self.is_nile() 437 } </pre>
omdurman-types/src/lib.rs:503 GH:omdurman-types/src/lib.rs:503	Terrain::is_crossroad	<pre> 501 502 /// Whether roads converge at this hex's centre. 503 pub fn is_crossroad(self) -> bool { 504 matches!(self.road(), Road::Crossroad) 505 } </pre>
omdurman-hexmap/src/map.rs:17 GH:omdurman-hexmap/src/map.rs:17	GameMap::roads	<pre> 15 pub hexes: HashMap<HexCoord, HexData>, 16 pub hexsides: HashMap<HexsideRef, HexsideKind>, 17 pub roads: HashSet<HexsideRef>, 18 pub excluded: HashSet<HexCoord>, 19 pub overlay: OverlayParams, </pre>

Proven by: [omdurman-rules::src::terrain_chart::movement_column_matches_the_printed_chart](#)
[omdurman-types::src::lib::terrain_road_state_round_trips](#)

Covered by tests: [omdurman-rules::src::terrain_chart::clear_terrain_no_bonus](#) [omdurman-rules::src::terrain_chart::nile_is_impassable](#)
[omdurman-rules::src::terrain_chart::rough_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::swamp_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::hilltop_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::huts_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::movement_cost_convenience_matches_chart](#)
[omdurman-rules::src::terrain_chart::movement_cost_with_road_always_one](#)
[omdurman-rules::src::effects::tests::land_unit_may_not_enter_nile](#)
[omdurman-rules::src::terrain_chart::movement_cost_without_road_matches_terrain](#)

omdurman-rules::src::effects::tests::movement_cost_for_uses_terrain
omdurman-rules::src::effects::tests::movement_cost_for_road_costs_one
omdurman-rules::src::terrain_chart::road_gives_crossroad omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds
omdurman-rules::src::terrain_chart::terrain_chart_road_always_costs_one

§5.12 – Move up to allowance, hex by hex (cumulative MP cap)

implemented

manual page unknown

A player may move as many or as few of his units as desired during each movement phase, limited only by the units' movement allowance, the terrain costs paid in moving from hex to hex, and enemy zones of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:779 GH:omdurman-rules/src/effects/state.rs:779	can_move_unit	<pre>777 /// the same `RuleError` the `MoveUnit` effect 778 /// would on rejection. Lets the 779 /// UI gate input without mutating or duplicating 780 /// the rules. 781 pub fn can_move_unit(&self, unit_id: UnitId, cost: MovementPoints) -> Result<(), RuleError> { 782 self.can_move_unit_to(unit_id, None, cost) 783 } 784 }</pre>
omdurman-rules/src/effects/state.rs:1474 GH:omdurman-rules/src/effects/state.rs:1474	mp_spent	<pre>1472 1473 /// Movement points `unit_id` has already spent 1474 /// this turn (§5.11/§5.12). 1475 pub fn mp_spent(&self, unit_id: UnitId) -> i16 { 1476 self.mp_spent_this_turn.get(&unit_id).copied().unwrap_or(0) 1477 } 1478 }</pre>

Covered by tests: omdurman-rules::src::effects::tests::cumulative_move_cost_may_not_exceed_allowance
omdurman-rules::src::effects::tests::wrong_faction_move_is_rejected
omdurman-rules::src::effects::tests::wrong_faction_gunboat_move_is_rejected

§5.13 – No MP accumulation between turns

implemented

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A unit may never accumulate movement points from turn to turn, nor may a unit transfer unused movement points to other units. A unit's unused movement points in any given turn are considered lost.

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:204 GH:omdurman-rules/src/effects/dispatch.rs:204	end_player_turn	<pre>202 203 /// End the current player's turn: recover disrupted 204 /// units, switch active player, advance turn index (rulebook 205 /// §4). 206 pub fn end_player_turn(state: &mut GameState) -> Result<(), RuleError> { 207 #[cfg(not(feature = "kani"))] 208 debug!(</pre>

Covered by tests: omdurman-rules::src::effects::tests::unused_movement_points_do_not_carry_over

§5.21 – Friendlies transport via gunboat

implemented

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In general, naval transport missions are not allowed, i.e. gunboats may not carry any land units. The sole exception is that the Anglo-Egyptian player may transport the surviving units of the “Friendlies” brigade from the east bank of the Nile to the west bank after, and only after, the Dervish east bank unit (Isa Zachneih) has been eliminated. The transport is accomplished in the following sequence: a) on any turn that a “Friendlies” unit and any Anglo-Egyptian gunboat start their turn adjacent, that unit may load onto (i.e. stack with) the gunboat; b) during the Anglo-Egyptian player’s next turn the gunboat may move to any Nile hex adjacent to a west bank hex (up to the gunboat’s movement allowance); c) on the Anglo-Egyptian player’s third turn the “Friendlies” unit may disembark and move normally, paying the normal terrain cost for the first hex entered. The gunboat may also move normally that turn.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:629 GH:omdurman-rules/src/lib.rs:629	is_friendlylies	<pre> 627 /// "Friendlies" units obey several special rules (\$5.21, \$5.23, \$6.52, 628 /// \$9.14 victory conditions). pub fn is_friendlylies(&self) -> bool { 629 matches!(630 self, 631 self, </pre>
omdurman-rules/src/effects/state.rs:2167 GH:omdurman-rules/src/effects/state.rs:2167	friendlylies_transport_off	<pre> 2165 /// regardless of selection. Pairs with [`GameEffect::FriendlyliesTransport`] 2166 /// so the UI can offer exactly the action the engine would accept. 2167 pub fn friendlylies_transport_offer(&self, selected: Option<UnitId>) -> Option<FriendlyliesAction> { 2168 match self.friendlylies_transport { 2169 None => { </pre>
omdurman-rules/src/lib.rs:829 GH:omdurman-rules/src/lib.rs:829	loaded_on	<pre> 827 pub disrupted: bool, 828 /// `Some(gunboat)` after a "Friendlies" unit loads onto a gunboat (\$5.21). 829 pub loaded_on: Option<UnitId>, 830 /// Set while the unit is building Zariba hexsides -- neither offensive 831 /// fire nor melee allowed that turn (\$5.3). </pre>
omdurman-rules/src/effects/effect.rs:196 GH:omdurman-rules/src/effects/effect.rs:196	FriendlyliesTransport	<pre> 194 195 /// Load/disembark the "Friendlies" brigade via gunboat (rulebook \$5.21). 196 FriendlyliesTransport(crate::FriendlyliesAction), 197 198 // -- Optional rules ----- </pre>
omdurman-rules/src/effects/river.rs:12 GH:omdurman-rules/src/effects/river.rs:12	apply_friendlylies_transpo	<pre> 10 /// unit is freed (a disembarking `MoveUnit` should follow, costed by 11 /// terrain). 12 pub fn apply_friendlylies_transport(13 state: &mut GameState, 14 action: FriendlyliesAction, </pre>
omdurman-rules/src/lib.rs:1077 GH:omdurman-rules/src/lib.rs:1077	FriendlyliesAction	<pre> 1075 /// tracks each unit-gunboat pair independently. 1076 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1077 pub enum FriendlyliesAction { 1078 /// Turn N (the load turn): unit and gunboat started adjacent; unit 1079 /// loads onto (stacks with) the gunboat. </pre>
omdurman-rules/src/lib.rs:1097 GH:omdurman-rules/src/lib.rs:1097	TransportState	<pre> 1095 /// third turn. 1096 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1097 pub enum TransportState { 1098 /// Turn N (the load turn): unit and gunboat started adjacent; unit 1099 /// loads onto (stacks with) the gunboat. </pre>

Covered by tests: [omdurman-rules::src::effects::tests::friendlies_transport_offer_load_requires_prerequisites](#)
[omdurman-rules::src::effects::tests::friendlies_transport_offer_follows_state_machine](#)

§5.22 – Land units may never enter a Nile River hex

implemented

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With the exception of 5.21, land units may never enter a Nile River hex. Only gunboats may enter and move along Nile River hexes.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:796 GH:omdurman-rules/src/effects/state.rs:796	can_move_unit_to	<pre>794 /// 795 /// [<code>`hex_in_enemy_zoc`</code>]: Self::hex_in_enemy_zoc 796 pub fn can_move_unit_to(797 &self, 798 unit_id: UnitId,</pre>
omdurman-rules/src/effects/state.rs:468 GH:omdurman-rules/src/effects/state.rs:468	in_deployment_zone	<pre>466 /// plan / UI rather than this hex predicate. /// Documented, not silently 467 /// dropped. 468 pub fn in_deployment_zone(&self, player: Player, hex: HexCoord, is_boat: bool) -> bool { 469 // No board attached -> permissive (unit tests, unbound session). 470 if self.board.terrain.is_empty() {</pre>

Covered by tests: [omdurman-rules::src::effects::tests::campaign_deployment_is_boat_land_exclusive](#)
[omdurman-rules::src::effects::tests::fok_ae_gunboat_deploys_only_on_nile](#)
[omdurman-rules::src::effects::tests::fok_ae_land_unit_rejected_on_nile](#)
[omdurman-rules::src::effects::tests::deploy_via_real_sprite_resolution_matches_engine](#)
[omdurman-rules::src::effects::tests::fok_dervish_land_unit_rejected_on_nile](#)
[omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_nile](#)

§5.23 – Walled city entry restrictions

implemented

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Only certain units may enter the walled portion of Omdurman. For the Dervish player these are the Khalifa unit, the three Dervish artillery units, and the Taiasha units (the Khalifa's bodyguard). Any Anglo-Egyptian units that can get to the walled city may enter it (except gunboats and "Friendlies"). Units entering and/or exiting the walled city may only do so through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:152 GH:omdurman-types/src/lib.rs:152	HexsideRef	<pre>150 /// data by [<code>`HexsideRef`</code>]. 151 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 152 pub struct HexsideRef { 153 pub a: HexCoord, 154 pub b: HexCoord,</pre>
omdurman-types/src/lib.rs:187 GH:omdurman-types/src/lib.rs:187	HexsideKind	<pre>185 strum::EnumIter, 186 }] 187 pub enum HexsideKind { 188 /// City wall (Khartoum, walled city of Omdurman). Blocks LOS, blocks 189 /// movement except across gates/breaches (§5.23), blocks ZOC into the city</pre>

omdurman-types/src/lib.rs:258 GH:omdurman-types/src/lib.rs:258	blocks_movement	<pre> 256 /// `omdurman-rules`). The trench *end* variants are therefore intentionally 257 /// not blocking. 258 pub fn blocks_movement(self) -> bool { 259 matches!(260 self, </pre>
omdurman-rules/src/lib.rs:655 GH:omdurman-rules/src/lib.rs:655	may_enter_walled_city	<pre> 653 /// Taiasha bodyguard may enter. Anglo-Egyptian: any unit that can reach the 654 /// walled city *except* gunboats and "Friendlies". 655 pub fn may_enter_walled_city(&self) -> bool { 656 match self { 657 // §5.23 Dervish: Khalifa, artillery, Taiasha. </pre>
omdurman-rules/src/board.rs:250 GH:omdurman-rules/src/board.rs:250	is_walled_city	<pre> 248 /// are its seeds). The set is derived once from the board data, replacing 249 /// the older "at least two of six hexsides are Wall/Gate/Breach" heuristic. 250 pub fn is_walled_city(&self, hex: HexCoord) -> bool { 251 // Membership in the precomputed enclosed area (see `walled_city`). 252 // Palace/Tomb hexes are always part of it (they are the seeds). </pre>
omdurman-rules/src/effects/error.rs:90 GH:omdurman-rules/src/effects/error.rs:90	WalledCityEntry	<pre> 88 89 #[error("unit {0:?} is not eligible to enter the walled city of Omdurman at {1:?} (§5.23)")] 90 WalledCityEntry(UnitId, HexCoord), 91 92 #[error("movement cost {cost:?} exceeds allowance {allowance:?}")] </pre>

Proven

by:

[omdurman-types::src::lib::hexside_ref_is_order_independent](#) [omdurman-types::src::lib::hexside_blocking_classifiers_are_exact](#)

Covered

by

tests:

[omdurman-rules::src::effects::tests::can_move_rejects_wall_hexside](#)
[omdurman-rules::src::effects::tests::can_move_allows_gate_hexside](#)
[omdurman-rules::src::effects::tests::walled_city_entry_allows_khalifa](#)
[omdurman-rules::src::effects::tests::walled_city_entry_rejects_unauthorized_dervish](#)
[omdurman-rules::src::effects::tests::walled_city_entry_rejects_ae_gunboat](#)
[omdurman-rules::src::effects::tests::walled_city_entry_not_enforced_for_fok](#)
[omdurman-rules::src::board_data::campaign_walled_city_is_enclosed_by_walls](#)
[omdurman-rules::src::effects::tests::breach_is_an_opening_for_los_and_effective_kinds](#)

§5.24 – Gunboat upstream/downstream movement

implemented

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Note that gunboats have two movement allowances separated by a slash, e.g. 10/16. The smaller number is the movement allowance when moving upstream, i.e. against the current (the direction of the current is indicated by arrows in the Nile). The larger number is the movement allowance when moving downstream, i.e. with the current. Gunboats may combine movement in both directions, but if they move even one hex upstream, their upstream movement allowance is their maximum movement allowance for that turn, and may not be exceeded.

(see manual for full text)

File	Symbol	Code Snippet
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omdurman-rules/src/lib.rs:552 GH:omdurman-rules/src/lib.rs:552	GunboatMovement	<pre> 550 /// the turn. 551 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 552 pub struct GunboatMovement { 553 pub upstream: MovementAllowance, 554 pub downstream: MovementAllowance, </pre>
omdurman-types/src/lib.rs:967 GH:omdurman-types/src/lib.rs:967	is_boat	<pre> 965 966 /// Gunboats use the split upstream/downstream movement allowance (§5.24). 967 pub fn is_boat(self) -> bool { 968 matches!(self, UnitKind::Gunboat { .. }) 969 } </pre>

Proven by: [omdurman-types::src::lib::unit_kind_fire_and_movement_capability_law_is_exact](#)
[omdurman-types::src::lib::hex_direction_from_index_is_total_mod_six](#)
[omdurman-types::src::lib::nile_flow_rotation_composes_and_never_leaves_the_nile](#)

Covered by tests: [omdurman-rules::src::unit_profiles::boat_annotation_yields_split_gunboat_movement](#)
[omdurman-rules::src::effects::tests::gunboat_upstream_cap_is_sticky_across_moves](#)

§5.25 – Dervish forts may not move

implemented

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Dervish forts may not move in any way once placed.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:813 GH:omdurman-rules/src/lib.rs:813	Immobile	<pre> 811 Gunboat(GunboatMovement), 812 /// Forts may not move once placed (§5.25). 813 Immobile, 814 } 815 </pre>
omdurman-types/src/lib.rs:924 GH:omdurman-types/src/lib.rs:924	UnitKind::Fort	<pre> 922 /// Permanent emplacement (§6.54): fire (artillery) / melee (defensive). 923 /// May not move once placed (§5.25). 924 Fort { fire: i32, melee: i32 }, 925 /// Dervish leader (§6.51): fire / melee / movement. May melee attack (§7.4). 926 DervishLeader { </pre>
omdurman-rules/src/lib.rs:813 GH:omdurman-rules/src/lib.rs:813	UnitMovement::Immobile	<pre> 811 Gunboat(GunboatMovement), 812 /// Forts may not move once placed (§5.25). 813 Immobile, 814 } 815 </pre>

Covered by tests: [omdurman-rules::src::effects::tests::forts_are_never_advance_eligible](#)

§5.26 – Units stop on entering enemy ZOC

implemented

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Units must stop their movement immediately upon entering an enemy zone of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
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omdurman-rules/src/effects/state.rs:796 GH:omdurman-rules/src/effects/state.rs:796	can_move_unit_to	794 795 796 797 798	/// /// [`hex_in_enemy_zoc`]: Self::hex_in_enemy_zoc pub fn can_move_unit_to(&self, unit_id: UnitId,
omdurman-rules/src/effects/state.rs:1600 GH:omdurman-rules/src/effects/state.rs:1600	hex_in_enemy_zoc	1598 1599 1600 1601 1602	/// does not extend into or out of a Nile hex. With no board loaded these /// reduce to the plain adjacency rule. pub fn hex_in_enemy_zoc(&self, hex: HexCoord,

Covered by tests: omdurman-rules::src::effects::tests::unit_entering_enemy_zoc_may_move_no_further_that_turn
omdurman-rules::src::effects::tests::zoc_transit_check_uses_the_actual_path

§5.41 – All units except AE leaders exert ZOC

implemented

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All units except Anglo-Egyptian leaders exert a zone of control (hereafter called a ZOC) into their six adjacent hexes (exception: Gunboats exert a ZOC only against enemy gunboats). Disrupted units have no ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:1854 GH:omdurman-rules/src/effects/state.rs:1854	unit_projects_zoc_rule	<pre>1852 /// * A fort projects ZOC out of its hex even when unoccupied (§5.44), 1853 /// modelled by the fort unit projecting normally. 1854 pub fn unit_projects_zoc_rule(1855 unit: &UnitPlacement, 1856 mover_player: Player,</pre>
omdurman-rules/src/lib.rs:868 GH:omdurman-rules/src/lib.rs:868	ZocReason	<pre>866 /// Used by the engine when answering "is this hex in an enemy ZOC?". 867 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 868 pub enum ZocReason { 869 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 870 /// leader (§5.41) projects ZOC into each of its six adjacent hexes.</pre>
omdurman-rules/src/effects/state.rs:1582 GH:omdurman-rules/src/effects/state.rs:1582	unit_projects_zoc	<pre>1580 /// §5.44) need the game map, which the engine does not hold; the app layers 1581 /// those on top. This is the position/kind/ disruption core of the rule. 1582 pub fn unit_projects_zoc(1583 &self, 1584 unit: &UnitPlacement,</pre>
omdurman-rules/src/effects/state.rs:1639 GH:omdurman-rules/src/effects/state.rs:1639	zoc_hexes	<pre>1637 /// ZOC covers a given hex; this function returns *which* hexes a 1638 /// specific unit covers. 1639 pub fn zoc_hexes(1640 &self, 1641 unit: &UnitPlacement,</pre>

Proven by: omdurman-rules::src::effects::unit_projects_zoc_matches_manual_clauses
omdurman-types::src::lib::player_opponent_is_a_fixed_point_free_involution

Covered by tests: omdurman-rules::src::effects::tests::zoc_hexes_empty_for_anglo_egyptian_leader
omdurman-rules::src::effects::tests::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions
omdurman-rules::src::effects::tests::zoc_hexes_empty_for_disrupted_unit
omdurman-rules::src::effects::tests::zoc_hexes_matches_hex_in_enemy_zoc omdurman-rules::src::effects::tests::zoc_hexes_excludes_nile
omdurman-rules::src::effects::tests::zoc_hexes_excludes_khor

§5.42 – No MP cost to enter/leave enemy ZOC

implemented

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There is no movement point cost to enter or leave an enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:966 GH:omdurman-rules/src/effects/state.rs:966	movement_cost_for	<pre>964 /// 965 /// §5.42: entering or leaving an enemy ZOC adds no MP cost. 966 pub fn movement_cost_for(967 &self, 968 unit: &UnitPlacement,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::entering_enemy_zoc_costs_no_extra_mp](#)

§5.43 – Units stop when entering enemy ZOC

implemented

manual page unknown

All units must stop when they enter an enemy ZOC and may move no further that turn. In their next movement phase they may withdraw or, if desired, move directly into another enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:796 GH:omdurman-rules/src/effects/state.rs:796	can_move_unit_to	<pre>794 /// 795 /// [`hex_in_enemy_zoc`]: Self::hex_in_enemy_zoc 796 pub fn can_move_unit_to(797 &self, 798 unit_id: UnitId,</pre>
omdurman-rules/src/effects/state.rs:1600 GH:omdurman-rules/src/effects/state.rs:1600	hex_in_enemy_zoc	<pre>1598 /// does not extend into or out of a Nile hex. With no board loaded these 1599 /// reduce to the plain adjacency rule. 1600 pub fn hex_in_enemy_zoc(1601 &self, 1602 hex: HexCoord,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::unit_entering_enemy_zoc_may_move_no_further_that_turn](#)

§5.44 – ZOC limitations (walls, khor, fort, Nile, Zariba)

implemented

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ZOCs do not extend into or out of a Nile River hex (exception: Gunboats, see 5.41). ZOCs do not extend across a khor, into a fort, or into a hex inside the walled city across a wall hexside. ZOCs do extend out of a fort (even if unoccupied), and from a walled city hex into an adjacent non-walled-city hex across a wall hexside. ZOCs also extend out of (but not into) a walled city hex across a gate hexside. ZOCs extend both ways across a breach hexside. ZOCs also extend out of, but not into, a hut or building hex. In the historical scenario ZOCs extend out of, but not into, the Zariba across a Zariba hexside (also in the campaign game if the Zariba is constructed).

(see manual for full text)

See also: §5.41

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:868 GH:omdurman-rules/src/lib.rs:868	ZocReason	<pre>866 /// Used by the engine when answering "is this hex in 867 an enemy ZOC?". 867 #[derive(Serialize, Deserialize, Clone, Copy, 868 PartialEq, Eq, Debug)] 868 pub enum ZocReason { 869 /// Normal ZOC: any non-disrupted unit other than 870 an Anglo-Egyptian 870 /// leader (§5.41) projects ZOC into each of its 870 six adjacent hexes.</pre>
omdurman-types/src/lib.rs:192 GH:omdurman-types/src/lib.rs:192	Wall	<pre>190 /// (§5.44), blocks melee (§7.2), blocks advance- 191 after-combat (§6.82). 191 #[default] 192 Wall, 193 /// Gate hexside in a wall. ZOC extends *out of* 194 the walled city through 194 /// gates but not into it (§5.44). Melee may be 194 made through a gate (§7.2).</pre>
omdurman-types/src/lib.rs:201 GH:omdurman-types/src/lib.rs:201	Khor	<pre>199 /// Khor -- gully/wadi. ZOCs do not extend across 200 (§5.44); advance after 201 /// combat may not cross (§6.82). 201 Khor, 202 /// Crest line. Blocks LOS unless the firer is on 203 the higher side 203 /// (§6.3 note 7).</pre>
omdurman-rules/src/lib.rs:881 GH:omdurman-rules/src/lib.rs:881	ZocReason::Zariba	<pre>879 /// Zariba hexside ZOC behaviour in the historical 880 scenario / when the 881 /// Zariba is constructed (§5.44). 881 Zariba, 882 } 883 </pre>
omdurman-rules/src/effects/state.rs:1582 GH:omdurman-rules/src/effects/state.rs:1582	unit_projects_zoc	<pre>1580 /// §5.44) need the game map, which the engine 1581 does not hold; the app layers 1581 /// those on top. This is the position/kind/ 1582 disruption core of the rule. 1582 pub fn unit_projects_zoc(1583 &self, 1584 unit: &UnitPlacement,</pre>
omdurman-types/src/lib.rs:275 GH:omdurman-types/src/lib.rs:275	HexsideKind::blocks_zoc	<pre>273 /// cannot express; those are left to the caller. 274 This predicate captures the 275 /// symmetric "does not extend across" cases. 275 pub fn blocks_zoc(self) -> bool { 276 matches!(277 self,</pre>
omdurman-rules/src/effects/state.rs:1600 GH:omdurman-rules/src/effects/state.rs:1600	hex_in_enemy_zoc	<pre>1598 /// does not extend into or out of a Nile hex. 1599 With no board loaded these 1599 /// reduce to the plain adjacency rule. 1600 pub fn hex_in_enemy_zoc(1601 &self, 1602 hex: HexCoord,</pre>

Proven by: [omdurman-rules::src::effects::unit_projects_zoc_matches_manual_clauses](#)
[omdurman-types::src::lib::hexside_blocking_classifiers_are_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::zoc_hexes_excludes_nile](#) [omdurman-rules::src::effects::tests::zoc_hexes_excludes_khor](#)
[omdurman-rules::src::effects::tests::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions](#)

§5.51 – Stacking limit (4 units + leaders, gunboats isolated)

implemented

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No more than four units may occupy a hex, with the exception of leaders and gunboats. All leader units are free stacking, i.e. they may stack in addition to the four-unit-per-hex stacking limitation. Gunboats may not stack with any other unit (Exception: 5.21). Players may move through friendly units at no additional cost in movement points. The stacking limitation applies only at the end of the movement phase and during combat.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:893 GH:omdurman-rules/src/lib.rs:893	OverLimit	<pre> 891 /// (§6.51: a Dervish unit occupying his hex eliminates him). 892 #[error("hex stack exceeds the four-unit limit [\$5.51]")] 893 OverLimit, 894 /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21). 895 #[error("gunboats may not stack with non-gunboat units [\$5.51]")] </pre>
omdurman-rules/src/lib.rs:896 GH:omdurman-rules/src/lib.rs:896	GunboatStack	<pre> 894 /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21). 895 #[error("gunboats may not stack with non-gunboat units [\$5.51]")] 896 GunboatStack, 897 /// "Units of different Dervish tribes may not stack together" (§5.52). 898 /// The Dervish artillery (§9.322's three guns) is not a tribe but is not </pre>
omdurman-rules/src/lib.rs:906 GH:omdurman-rules/src/lib.rs:906	EnemyCohabitation	<pre> 904 /// lone Anglo-Egyptian leader is exempt (§6.51). 905 #[error("enemy units may not share a hex; melee, not movement, engages them [\$5.51, §7.1]")] 906 EnemyCohabitation, 907 /// "If Dervish leaders elect to stack, they may only stack with units of 908 /// their command (i.e. colour)" (§5.53). </pre>
omdurman-rules/src/effects/state.rs:1533 GH:omdurman-rules/src/effects/state.rs:1533	check_stacking	<pre> 1531 /// Dervish artillery is its own group) may not stack together. 1532 /// * §5.53 -- a Dervish leader may stack only with units of its command. 1533 pub fn check_stacking(1534 &self, 1535 mover: &UnitPlacement, </pre>
omdurman-rules/src/effects/state.rs:1765 GH:omdurman-rules/src/effects/state.rs:1765	stacking_rule	<pre> 1763 /// artillery as its own group) may not share a hex. 1764 /// * §5.53: a Dervish leader stacks only with units of its command. 1765 pub fn stacking_rule(occupants: &[&UnitPlacement]) -> Result<(), crate::StackingError> { 1766 use crate::StackingError; 1767 </pre>

Proven by:

- [omdurman-rules::src::effects::stacking_rule_cohabitation_is_exact](#)
- [omdurman-rules::src::effects::stacking_rule_is_symmetric](#)
- [omdurman-rules::src::effects::stacking_rule_leaders_are_free_stacking](#)
- [omdurman-rules::src::effects::stacking_rule_limit_is_four_counted_units](#)
- [omdurman-rules::src::effects::stacking_rule_gunboat_never_shares](#)

Covered by tests:

- [omdurman-rules::src::effects::tests::stacking_over_limit_rejected](#)
- [omdurman-rules::src::effects::tests::mid_move_stacking_allows_pass_through](#)
- [omdurman-rules::src::effects::tests::mid_move_stacking_rejects_over_limit_destination](#)
- [omdurman-rules::src::effects::tests::validate_stacking_invariants_clean_state](#)
- [omdurman-rules::src::effects::tests::validate_stacking_invariants_catches_stacking_violation](#)
- [omdurman-rules::src::effects::tests::validate_stacking_invariants_allows_leaders_stacking](#)
- [omdurman-rules::src::effects::tests::deploy_rejects_enemy_cohabitation_during_setup](#)
- [omdurman-rules::src::effects::tests::validate_stacking_invariants_catches_enemy_cohabitation](#)
- [omdurman-rules::src::effects::tests::ae_garrison_stacks_freely_under_gordon](#)
- [omdurman-rules::src::effects::tests::melee_mandatory_advance_leaders_are_free_stacking](#)

§5.52 – Different Dervish tribes may not stack together

implemented

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The units of different Dervish tribes may not stack together, even if they are the same color (e.g. although both are green, Mulazmin and Jehadia units may not stack with each other).

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:902 GH:omdurman-rules/src/lib.rs:902	DervishTribeMix	<pre>900 /// [`UnitIdentity::dervish_stacking_group`]). 901 #[error("Dervish units of different tribes may not stack [\$5.52]")] 902 DervishTribeMix, 903 /// A unit may never share a hex with enemy units (\$5.51, \$7.1); only the 904 /// lone Anglo-Egyptian leader is exempt (\$6.51).</pre>
omdurman-rules/src/lib.rs:707 GH:omdurman-rules/src/lib.rs:707	dervish_stacking_group	<pre>705 /// `None` for every unit the §5.52 law does not constrain: Dervish 706 /// leaders, forts and gunboats, and all Anglo- Egyptian units. 707 pub fn dervish_stacking_group(&self) -> Option<DervishStackingGroup> { 708 match self { 709 UnitIdentity::DervishTribal { tribe } => Some(DervishStackingGroup::Tribe(*tribe)),</pre>

Proven by: [omdurman-rules::src::effects::stacking_rule_group_purity_is_exact](#)

Covered by tests: [omdurman-rules::src::unit_profiles::green_sections_are_mulazmin_tribal_units](#)
[omdurman-rules::src::effects::tests::deploy_rejects_dervish_tribe_mix](#)
[omdurman-rules::src::effects::tests::deploy_rejects_hadendowa_on_dervish_gun](#)
[omdurman-rules::src::effects::tests::dervish_guns_stack_with_guns_and_their_leader](#)
[omdurman-rules::src::effects::tests::validate_stacking_invariants_clean_state](#)

§5.53 – Leader stacking with command colour only

implemented

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Leader units are not required to stack. If Dervish leaders elect to stack, however, they may only stack with units of their command (i.e. color). For example, Sheik El Din may only stack with Mulazmins or Jehadias.

****5.54) Anglo-Egyptian Brigade Integrity:**** All British, Sudanese, and Egyptian infantry units have their brigade designation printed in the upper right corner (e.g. “2B” = 2nd British Brigade; “3E” = 3rd Egyptian Brigade, etc.). In any combat phase in which all four infantry battalions belonging to any Anglo-Egyptian infantry brigade are stacked in the same hex they are said to have brigade integrity. Stacks having brigade integrity receive a +1 modifier to their fire combat die roll provided they all fire at the same enemy occupied hex. This modifier is in addition to the normal +1 bonus given to all Anglo-Egyptian direct fire attacks (see 6.24).

(see manual for full text)

See also: §5.54, §6.24

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:910 GH:omdurman-rules/src/lib.rs:910	DervishLeaderCommandMismatch	<pre>908 /// their command (i.e. colour)" (§5.53). 909 #[error("Dervish leader may only stack with units of their own command [\$5.53]")] 910 DervishLeaderCommandMismatch, 911 } 912 </pre>

Proven by: [omdurman-rules::src::effects::stacking_rule_leaders_are_free_stacking](#)

Covered by tests: [omdurman-rules::src::effects::tests::dervish_leader_stacks_only_with_command_colour](#)
[omdurman-rules::src::unit_profiles::command_owns_unit_follows_tribes_and_brigades](#)

§5.54 – Anglo-Egyptian Brigade Integrity

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:916 GH:omdurman-rules/src/lib.rs:916	BrigadeIntegrity	<pre>914 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 915 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 916 pub enum BrigadeIntegrity { 917 None, 918 Integrated(BrigadeId),</pre>
omdurman-rules/src/lib.rs:771 GH:omdurman-rules/src/lib.rs:771	brigade_integrity	<pre>769 /// Only a full stack of four battalions qualifies. Three or fewer may still 770 /// stack and fire, but they receive no brigade- integrity bonus. 771 pub fn brigade_integrity(identities: &[UnitIdentity]) - > BrigadeIntegrity { 772 let Some(brigade) = identities.first().and_then(i i.brigade()) else { 773 return BrigadeIntegrity::None;</pre>
omdurman-rules/src/lib.rs:916 GH:omdurman-rules/src/lib.rs:916	FireModifier::BrigadeIntegrity	<pre>914 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 915 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 916 pub enum BrigadeIntegrity { 917 None, 918 Integrated(BrigadeId),</pre>
omdurman-rules/src/lib.rs:329 GH:omdurman-rules/src/lib.rs:329	BattalionOrdinal	<pre>327 /// brigade integrity requires all four stacked in one hex (§5.54). 328 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 329 pub enum BattalionOrdinal { 330 First = 1, 331 Second = 2,</pre>
omdurman-types/src/lib.rs:1043 GH:omdurman-types/src/lib.rs:1043	BrigadeId	<pre>1041 /// same field for uniform handling. 1042 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 1043 pub struct BrigadeId { 1044 pub number: u8, 1045 pub nationality: BrigadeNationality,</pre>

Proven by: [omdurman-rules::src::lib::brigade_integrity_requires_all_four_distinct_battalions_of_one_brigade](#)

Covered by tests: [omdurman-rules::src::unit_profiles::brigade_designation_ignored_for_non_infantry](#)
[omdurman-rules::src::unit_profiles::printed_brigade_designation_overrides_column](#)
[omdurman-rules::src::unit_profiles::section_owner_anglo_egyptian_sections](#)
[omdurman-rules::src::lib::brigade_integrity_four_battalions_returns_integrated](#)
[omdurman-rules::src::unit_profiles::ae_infantry_fourth_battalion_from_col_3](#)
[omdurman-rules::src::lib::brigade_integrity_empty_slice](#) [omdurman-rules::src::lib::brigade_integrity_friendlys_returns_none](#)
[omdurman-rules::src::lib::brigade_integrity_three_battalions_returns_none](#)
[omdurman-rules::src::lib::unit_identity_brigade_and_battalion_accessors](#)
[omdurman-rules::src::unit_profiles::ae_infantry_brigade_number_three_from_col_7](#)
[omdurman-rules::src::lib::brigade_integrity_non_infantry_returns_none](#)
[omdurman-rules::src::lib::brigade_integrity_mixed_brigades_returns_none](#)
[omdurman-rules::src::unit_profiles::ae_infantry_third_battalion_from_col_2](#)
[omdurman-rules::src::effects::tests::brigade_integrity_modifier_is_engine_derived](#)

 21/31 implemented

§6 – Fire Combat Phase

§6 – Fire Combat Phase

descriptive

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Fire Combat Phase

§6.1 – General Rules

descriptive

manual page unknown

General Rules

§6.2 – How To Have Fire Combat

descriptive

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How To Have Fire Combat

§6.3 – Line of Sight Table

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Line of Sight Table

This table is located on the back of this rulebook and should be self-explanatory. Locate the terrain type the firing unit is in and cross-index it with the terrain type the target unit is in. Terrain types in the intersecting box block line of sight, with exceptions as footnoted. Also study the “Special LOS Notes” given and remember that (with the exception of howitzer fire — see 6.64) you can’t fire at anything you can’t see!

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:57 GH:omdurman-rules/src/los_table.rs:57	LosLevel	<pre>55 serde::Serialize, serde::Deserialize, Clone, Copy, 56 PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 57)] 58 pub enum LosLevel { 59 Ground, 59 Rough,</pre>
omdurman-rules/src/los_table.rs:81 GH:omdurman-rules/src/los_table.rs:81	LosFeature	<pre>79 /// authored RON spellings. 80 #[derive(serde::Serialize, serde::Deserialize, Clone, 81 Copy, PartialEq, Eq, Hash, Debug)] 82 pub enum LosFeature { 83 /// A hex containing units (gunboats/forts excluded 83 Units,</pre>
omdurman-rules/src/los_table.rs:103 GH:omdurman-rules/src/los_table.rs:103	LosCondition	<pre>101 /// A positional condition from the LOS table Detail 102 footnotes. 103 #[derive(serde::Serialize, serde::Deserialize, Clone, 104 Copy, PartialEq, Eq, Hash, Debug)] 105 pub enum LosCondition { 106 /// (1) Blocks only if the ray passes through more</pre>

		than two such features. 105 MoreThanTwo,
omdurman-rules/src/los_table.rs:183 GH:omdurman-rules/src/los_table.rs:183	los_level	181 /// 182 /// For all other units, the level is derived from the terrain at `hex`. 183 pub fn los_level_for_unit(184 kind: UnitKind, 185 hex: HexCoord,
omdurman-rules/src/los_table.rs:183 GH:omdurman-rules/src/los_table.rs:183	los_level_for_unit	181 /// 182 /// For all other units, the level is derived from the terrain at `hex`. 183 pub fn los_level_for_unit(184 kind: UnitKind, 185 hex: HexCoord,
omdurman-rules/src/los_table.rs:223 GH:omdurman-rules/src/los_table.rs:223	blocking_rules	221 /// always blocks. Indexing is in-bounds by construction (both enums have 222 /// exactly three variants). 223 pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] { 224 crate::tables_data::LOS_CELLS[firer.index()][target.index()] 225 }
omdurman-rules/src/los_table.rs:558 GH:omdurman-rules/src/los_table.rs:558	has_los	556 /// board-only tools). 557 #[allow(clippy::too_many_arguments)] 558 pub fn has_los(559 board: &crate::board::BoardInfo, 560 from: HexCoord,
omdurman-types/src/lib.rs:230 GH:omdurman-types/src/lib.rs:230	HexsideKind::blocks_los	228 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 229 /// in `omdurman_rules::los_table`, not by this predicate. 230 pub fn blocks_los(self) -> bool { 231 matches!(self, HexsideKind::Wall HexsideKind::Crest) 232 }
omdurman-types/src/lib.rs:230 GH:omdurman-types/src/lib.rs:230	Terrain::blocks_los	228 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 229 /// in `omdurman_rules::los_table`, not by this predicate. 230 pub fn blocks_los(self) -> bool { 231 matches!(self, HexsideKind::Wall HexsideKind::Crest) 232 }
omdurman-types/src/lib.rs:452 GH:omdurman-types/src/lib.rs:452	Terrain::is_los_trees	450 /// (§6.3 note 1). Retained for compatibility; the full LOS engine 451 /// checks `Terrain::Trees` directly. 452 pub fn is_los_trees(self) -> bool { 453 matches!(self, Terrain::Trees { .. }) 454 }

Proven

by: [omdurman-types::src::lib::line_between_forms_a_connected_ray](#) [omdurman-types::src::lib::hexside_blocking_classifiers_are_exact](#)
[omdurman-rules::src::los_table::los_level_depends_only_on_the_ground](#)
[omdurman-rules::src::los_table::los_level_overrides_hold_for_gunboats_and_forts](#)
[omdurman-rules::src::los_table::blocking_grid_is_total_and_walls_block_ground_firers](#)

Covered by tests: [omdurman-rules::src::los_table::los_level_mapping](#) [omdurman-rules::src::los_table::blocking_rules_all_cells_covered](#)
[omdurman-rules::src::los_table::has_los_empty_board_is_clear](#) [omdurman-rules::src::los_table::has_los_adjacent_clear](#)
[omdurman-rules::src::los_table::has_los_howitzzer_bypasses](#)
[omdurman-rules::src::los_table::has_los_wall_hexside_blocks_ground_to_ground](#)
[omdurman-rules::src::los_table::has_los_gate_hexside_passes](#) [omdurman-rules::src::los_table::has_los_breach_hexside_passes](#)
[omdurman-rules::src::effects::tests::breach_is_an_opening_for_los_and_effective_kinds](#)
[omdurman-rules::src::los_table::has_los_rough_intervening_blocks_ground_to_ground](#)

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omdurman-rules::src::los_table::has_los_two_tree_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::has_los_three_tree_hexes_block_ground_to_ground
omdurman-rules::src::los_table::has_los_two_hut_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::has_los_three_hut_hexes_block_ground_to_ground
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_clear_no_units
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_blocked_by_hilltop_unit
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_not_blocked_by_ground_unit
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_lower_level_passes
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_same_level_blocks
omdurman-rules::src::los_table::has_los_rough_to_rough_hilltop_blocks
omdurman-rules::src::los_table::has_los_ground_to_hilltop_intervening_hilltop_blocks
omdurman-rules::src::los_table::has_los_building_blocks_like_huts_ground_to_ground
omdurman-rules::src::los_table::has_los_two_building_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::los_level_for_unit_gunboat_is_rough
omdurman-rules::src::los_table::los_level_for_unit_fort_is_ground
omdurman-rules::src::los_table::los_level_for_unit_walled_city_adj_wall_is_rough
omdurman-rules::src::los_table::gunboat_firer_uses_rough_row_not_ground
omdurman-rules::src::los_table::los_reflexive_all_terrains
omdurman-rules::src::los_table::los_reflexive_hilltop
omdurman-rules::src::los_table::los_symmetric_ground_to_ground_no_units
omdurman-rules::src::los_table::los_howitzers_always_has_los
omdurman-rules::src::los_table::los_howitzers_same_hex
omdurman-rules::src::los_table::los_blocking_rules_match_reference_table
omdurman-rules::src::los_table::los_ground_to_ground_features_block_as_expected

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§6.4 – Fire Combat Sequence

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Fire Combat Sequence

The sequence of fire combat resolution is the same for both defensive and offensive fire combat. During the Dervish player turn the Anglo-Egyptian player executes 6.41 AND 6.42 as defensive fire, after which the Dervish player executes 6.41 as offensive fire. During the Anglo-Egyptian player turn the Dervish player executes 6.41 as defensive fire, after which the Anglo-Egyptian player executes 6.41 AND 6.42 as offensive fire.

****6.41) Direct Fire Subphase (Dervish and Anglo-Egyptian players):**** The firing player must first allocate all of his fire attacks, combining his units’ direct fire combat factors in any manner he wishes. After all fire has been allocated, the firing player then resolves his attacks in any order he wishes.

****6.42) Maxim Second Fire and Howitzer Fire Subphase (Anglo-Egyptian player only):**** Anglo-Egyptian named gunboats may now fire their artillery factor as howitzer fire (see 6.64) and all Maxim guns may fire a second time. Once

(see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:302 GH:omdurman-rules/src/lib.rs:302	FireSubPhase	<pre> 300 301 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 302 pub enum FireSubPhase { 303 /// Direct fire (§6.41). Both sides participate in this sub-phase. 304 DirectFire, </pre>
omdurman-rules/src/effects/effect.rs:72 GH:omdurman-rules/src/effects/effect.rs:72	FireCombat	<pre> 70 /// - Firers marked as fired; target hex marked as fired-at. 71 /// - Victory points awarded for eliminations. 72 FireCombat { attack: FireAttack, roll: DieRoll }, 73 74 /// Resolve a howitzer bombardment (two rolls: CRT + impact scatter) </pre>

Covered by tests: omdurman-rules::src::effects::tests::turn_advances_through_phases

§6.5 – Special Unit Capabilities

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Special Unit Capabilities

****6.51) Leader Units:**** Dervish leader units have a fire factor, a melee factor, and a movement factor. They may thus attack, melee, and be eliminated like any other combat unit. Their special benefit is that they stack free.

Anglo-Egyptian leaders have a movement factor only. They are eliminated if a) they are alone in a hex when a Dervish unit occupies or passes through that hex, or b) if all of the combat units a leader is stacked with are eliminated in fire combat or melee. The special function of Anglo-Egyptian leaders is that at least one must survive to occupy the Mahdi’s tomb hex if it is to be taken from the Dervish player (see 9.14).

****6.52) Anglo-Egyptian “Friendlies” Brigade:**** These units represent native volunteers in the Anglo-Egyptian army. They fire rifles on the Dervish Range Effects Table and melee with the Dervish melee modifier. They may not enter the walled city of Omdurman (see 5.23). They may be transferred to the west bank (see 5.21).

(see manual for full text)

See also: §5.21, §5.23, §5.44, §6.51, §6.52, §6.53, §6.54, §6.62, §6.63, §7.6, §9.14

§6.6 – Special Artillery Capabilities

descriptive
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Special Artillery Capabilities

§6.7 – Defensive Fire

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Defensive Fire

In Defensive Fire phase, all of the non-moving player’s units may fire at any of the moving player’s units in range, within the limitations imposed by the rules of combat (see 6.1 to 6.6). There is no advance after combat as a result of defensive fires.

See also: §6.1, §6.6

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:2044 GH:omdurman-rules/src/effects/state.rs:2044	<code>can_advance_after_comb</code>	<pre>2042 /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 2043 /// Wall/khor hexside restrictions are not enforced (no hexside map data). 2044 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 2045 let unit = self.unit_or_err(unit_id)?; 2046 // §6.7: there is no advance after combat as a result of defensive fire.</pre>

Covered by tests: `omdurman-rules::src::effects::tests::no_advance_after_defensive_fire`
`omdurman-rules::src::effects::tests::defensive_fire_opens_no_advance_window`

§6.8 – Offensive Fire

descriptive
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Offensive Fire

§6.11 – Fire combat factor printed on units

implemented

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The fire combat factor of the various unit types is printed directly on the units and is a numerical expression of the unit's fire strength.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:86 GH:omdurman-rules/src/lib.rs:86	FireFactor	<pre>84 /// Every possible value from the annotated counter 85 set is a named variant. 86 #[derive(Serialize, Deserialize, Clone, Copy, 87 PartialEq, Eq, Hash, Debug, strum::Display)] 88 pub enum FireFactor { 89 One = 1, 90 Three = 3,</pre>

Covered by tests: [omdurman-rules::src::lib::fire_factor_sum_to_row](#)

§6.12 – Fire combat is always voluntary

implicit

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Fire combat is always voluntary. A unit is never required to fire at enemy units merely because they are in range or adjacent.

§6.13 – Fire factor is unitary (may not be divided)

implicit

manual page unknown

If a unit elects to fire, its fire combat factor at an enemy unit, that fire combat factor is unitary. A unit's fire combat factor may not be divided up to fire at enemy units on different hexes.

§6.14 – Players may combine fire factors into one attack

implemented

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Players may combine fire during fire combat phase, i.e. they may fire at an enemy-occupied hex with as many friendly units as may legally do so, combining all of their fire combat factors into one attack. Note that in any given fire combat phase, however, a combat unit may only fire once and may only be fired at once (exceptions: Maxim guns and gunboats — see 6.4).

See also: §6.4

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:100 GH:omdurman-rules/src/lib.rs:100	sum_to_row	<pre>98 impl FireFactor { 99 /// Sum multiple fire factors and return the 100 corresponding Combat Results Table row (rulebook §6.11). 101 pub fn sum_to_row<'a>(factors: impl 102 IntoIterator<Item = &'a FireFactor>) -> FireFactorRow { 103 let total: u16 = factors.into_iter().map(f 104 f.value()).sum(); 105 crate::combat_results_table::FireFactorRow::from_total(total)</pre>

Covered by tests: `omdurman-rules::src::effects::tests::unit_may_only_be_fired_at_once_per_phase`
`omdurman-rules::src::effects::tests::gunboat_and_maxim_may_be_fired_at_repeatedly`
`omdurman-rules::src::effects::tests::rejected_fire_attack_does_not_mark_firers_as_fired`

§6.15 – May divide a stack to fire at different hexes

implicit

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Players may also divide a stack of units in order to fire at different enemy-occupied hexes. Anglo-Egyptian infantry units having brigade integrity, however, do not receive their +1 direct fire modifier unless they all fire at the same enemy-occupied hex (see 5.54).

See also: §5.54

§6.16 – Halving fire strength rounds down, minimum 1

implemented

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When halving fire combat strength, always round down each individual unit. For example, an Egyptian brigade of four battalions, each having a printed strength of 9 fire factors, will fire a total of 16 factors when halved. However, a unit's firing strength is never reduced below one by halving.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:519 GH:omdurman-rules/src/lib.rs:519	RangeBand	<pre>517 /// multiplied at a given distance (§6.22). 518 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 519 pub enum RangeBand { 520 Tripled, 521 Doubled,</pre>

Proven by: `omdurman-rules::src::lib::range_band_halved_is_max_of_one_and_floor_half`
`omdurman-rules::src::lib::range_band_apply_is_monotone_in_raw` `omdurman-rules::src::lib::disrupt_half_is_rounded_up`

Covered by tests: `omdurman-rules::src::lib::halving_rounds_down_and_never_below_one`

§6.21 – First check LOS before firing

implemented

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When combat units wish to fire at enemy units, first check the Line of Sight Table to be sure the firing unit can see the target hex (exception: howitzer fire, see 6.64).

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:223 GH:omdurman-rules/src/los_table.rs:223	blocking_rules	<pre>221 /// always blocks. Indexing is in-bounds by construction (both enums have 222 /// exactly three variants). 223 pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] { 224 crate::tables_data::LOS_CELLS[firer.index()] [target.index()] 225 }</pre>
omdurman-rules/src/los_table.rs:558 GH:omdurman-	has_los	<pre>556 /// board-only tools). 557 #[allow(clippy::too_many_arguments)] 558 pub fn has_los(</pre>

rules/src/ los_table.rs:558		559 board: &crate::board::BoardInfo, 560 from: HexCoord,	
--------------------------------	--	---	--

Covered by tests:
omdurman-rules::src::effects::tests::can_fire_at_rejects_blocked_los omdurman-rules::src::effects::tests::can_fire_at_allows_clear_los

§6.22 – Consult Range Effects Table

implemented

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Next consult the Range Effects Table to see if the firing unit's fire combat factor is tripled, doubled, normal, halved, or if the target hex is out of range. Add up the total number of fire combat factors firing at the enemy-occupied hex.

File	Symbol	Code Snippet
omdurman-rules/src/range_effects.rs:34 GH:omdurman-rules/src/range_effects.rs:34	ae_range_effects	32 /// Look up the range band for an Anglo-Egyptian weapon (§6.22, §6.24). 33 /// Distances > 10 are out of range for all weapons. 34 pub fn ae_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand { 35 band_at(faction_rows(true), weapon, distance) 36 }
omdurman-rules/src/range_effects.rs:40 GH:omdurman-rules/src/range_effects.rs:40	dervish_range_effects	38 /// Look up the range band for a Dervish weapon (§6.22). 39 /// Distances > 10 are out of range for all weapons. 40 pub fn dervish_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand { 41 band_at(faction_rows(false), weapon, distance) 42 }
omdurman-rules/src/lib.rs:519 GH:omdurman-rules/src/lib.rs:519	RangeBand	517 /// multiplied at a given distance (§6.22). 518 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 519 pub enum RangeBand { 520 Tripled, 521 Doubled,
omdurman-rules/src/lib.rs:187 GH:omdurman-rules/src/lib.rs:187	HexDistance	185 /// (rulebook §6.22, §7.5). 186 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 187 pub struct HexDistance(u16); 188 189 impl HexDistance {

Proven by:
omdurman-types::src::lib::adjacency_iff_distance_one omdurman-rules::src::lib::fire_factor_row_from_total_matches_printed_bands
omdurman-rules::src::lib::range_band_multiplier_arithmetic_is_exact
omdurman-rules::src::range_effects::range_effects_are_out_of_range_outside_the_printed_ten_hex_distance_window
omdurman-rules::src::range_effects::max_day_range_is_the_last_in_range_hex
omdurman-rules::src::effects::fire::range_band_for_routes_to_the_right_faction_table

Covered by tests:
omdurman-rules::src::range_effects::ae_rifles_doubled_at_range_1 omdurman-rules::src::range_effects::ae_rifles_halved_at_range_4
omdurman-rules::src::range_effects::ae_howitzer_range omdurman-rules::src::range_effects::dervish_rifles_shorter_range
omdurman-rules::src::range_effects::melee_only_range_1 omdurman-rules::src::range_effects::ae_range_effects_artillery_full
omdurman-rules::src::range_effects::ae_range_effects_maxims_match_rifles
omdurman-rules::src::range_effects::ae_range_effects_distance_over_10
omdurman-rules::src::range_effects::ae_range_effects_howitzer_halved_4_to_10
omdurman-rules::src::range_effects::dervish_range_effects_rifles omdurman-rules::src::range_effects::dervish_range_effects_artillery
omdurman-rules::src::range_effects::dervish_range_effects_maxims_and_howitzer
omdurman-rules::src::range_effects::dervish_range_effects_melee
omdurman-rules::src::range_effects::dervish_range_effects_distance_over_10
omdurman-rules::src::effects::tests::fire_combat_eliminate_target omdurman-rules::src::range_effects::max_day_range_all_combos

```

omdurman-rules::src::range_effects::range_effects_every_cell_ae_artillery
omdurman-rules::src::range_effects::range_effects_every_cell_ae_howitzer
omdurman-rules::src::range_effects::range_effects_every_cell_ae_maxims
omdurman-rules::src::range_effects::range_effects_every_cell_ae_rifles
omdurman-rules::src::range_effects::range_effects_every_cell_dervish_artillery
omdurman-rules::src::range_effects::range_effects_every_cell_dervish_maxims_howitzer
omdurman-rules::src::range_effects::range_effects_every_cell_dervish_rifles
omdurman-rules::src::range_effects::range_effects_every_cell_dervish_spears
omdurman-rules::src::effects::tests::can_fire_at_gates_phase_range_and_player
omdurman-rules::src::effects::tests::mixed_attack_bands_per_firer
omdurman-rules::src::range_effects::ae_range_effects_monotone_non_increasing
omdurman-rules::src::range_effects::ae_howitzer_has_minimum_range
omdurman-rules::src::range_effects::dervish_range_effects_monotone_non_increasing
omdurman-rules::src::range_effects::range_effects_first_range_max_effect_last_range_oor

```

§6.23 – Terrain defensive modifier

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Next check the Terrain Effects Chart to see if the enemy-occupied hex fired upon contains any terrain which gives the enemy units in that hex a defensive benefit. If so, apply this negative modifier to the roll of the ten-sided die and cross-index your net die roll on the Combat Results Table with the number of combat factors firing.

File	Symbol	Code Snippet
omdurman-rules/src/terrain_chart.rs:59 GH:omdurman-rules/src/terrain_chart.rs:59	defense_modifier	<pre> 57 58 /// Convenience: get the defense modifier for a terrain type (rulebook §6.23, Terrain Effects Chart). 59 pub fn defense_modifier(terrain: Terrain) -> i16 { 60 terrain_effects_chart(terrain).defense_modifier 61 } </pre>
omdurman-rules/src/lib.rs:937 GH:omdurman-rules/src/lib.rs:937	FireModifier::Terrain	<pre> 935 /// Negative modifier from the Terrain Effects Chart applied to the 936 /// defender's hex (§6.23). 937 Terrain(i16), 938 /// -2 thorn-hedge defensive modifier (§9.231). 939 ZaribaThornHedge, </pre>

Proven by: [omdurman-rules::src::lib::fire_modifier_keeps_roll_legal](#)
[omdurman-rules::src::terrain_chart::defence_column_never_helps_the_attacker](#)

Covered by tests: [omdurman-rules::src::terrain_chart::clear_terrain_no_bonus](#) [omdurman-rules::src::terrain_chart::building_gives_minus_3](#)
[omdurman-rules::src::terrain_chart::palm_grove_gives_minus_2](#) [omdurman-rules::src::terrain_chart::rough_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::swamp_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::hilltop_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::huts_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::defense_modifier_convenience_matches_chart](#)
[omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds](#)
[omdurman-rules::src::terrain_chart::terrain_defense_modifier_non_positive](#)

§6.24 – Anglo-Egyptian direct fire accuracy bonus and brigade integrity

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All Anglo-Egyptian direct fire attacks receive a +1 modifier to their die roll as an accuracy bonus. In addition, any stack of Anglo-Egyptian infantry having brigade integrity (see 5.54) receives a +1 modifier to their die roll if all four fire at the same enemy-occupied hex. These modifiers are cumulative.

See also: §5.54

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:931 GH:omdurman-rules/src/lib.rs:931	AngloEgyptianDirectFire	<pre> 929 pub enum FireModifier { 930 /// +1 to all Anglo-Egyptian *direct* fire (§6.24). 931 AngloEgyptianDirectFire, 932 /// +1 brigade integrity, applied only if all four battalions fire at 933 /// the same enemy-occupied hex (§5.54, §6.24). </pre>
omdurman-rules/src/lib.rs:916 GH:omdurman-rules/src/lib.rs:916	BrigadeIntegrity	<pre> 914 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 915 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 916 pub enum BrigadeIntegrity { 917 None, 918 Integrated(BrigadeId), </pre>
omdurman-rules/src/lib.rs:1031 GH:omdurman-rules/src/lib.rs:1031	FireModifier::die_modifier	<pre> 1029 impl MeleeModifier { 1030 /// Return the numeric die-roll modifier for this melee bonus/penalty (rulebook §7.7, §9.232). 1031 pub fn die_modifier(self) -> i16 { 1032 match self { 1033 MeleeModifier::DervishStandard => 2, </pre>

Proven by: omdurman-rules::src::lib::die_roll_apply_modifier_is_total

Covered by tests: omdurman-rules::src::effects::tests::brigade_integrity_modifier_is_engine_derived
omdurman-rules::src::effects::tests::fire_modifiers_are_engine_derived_and_mismatches_rejected

§6.41 – Direct Fire Subphase

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:304 GH:omdurman-rules/src/lib.rs:304	DirectFire	<pre> 302 pub enum FireSubPhase { 303 /// Direct fire (§6.41). Both sides participate in this sub-phase. 304 DirectFire, 305 /// Anglo-Egyptian only: Maxim second fire + named- gunboat howitzer fire (§6.42). 306 MaximSecondAndHowitzer, </pre>

Covered by tests: omdurman-rules::src::effects::tests::disrupted_unit_cannot_fire omdurman-rules::src::effects::tests::fire_combat_elimimates_target

§6.42 – Maxim Second Fire and Howitzer Fire Subphase

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:306 GH:omdurman-rules/src/lib.rs:306	MaximSecondAndHowitzer	<pre> 304 DirectFire, 305 /// Anglo-Egyptian only: Maxim second fire + named- gunboat howitzer fire (§6.42). 306 MaximSecondAndHowitzer, 307 } 308 </pre>
omdurman-types/src/lib.rs:985 GH:omdurman-types/src/lib.rs:985	fires_twice	<pre> 983 /// Maxim guns fire twice per turn -- once in the Direct Fire Subphase and 984 /// again in the Maxim Second Fire Subphase (rulebook §6.42). 985 pub fn fires_twice(self) -> bool { 986 matches!(self, UnitKind::Maxim { .. }) 987 } </pre>

Proven by: `omdurman-types::src::lib::unit_kind_fire_and_movement_capability_law_is_exact`

Covered by tests: `omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10`
`omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7`
`omdurman-rules::src::effects::tests::advance_window_bridges_fire_subphase_and_closes_at_melee`
`omdurman-rules::src::effects::tests::fired_at_tracker_resets_at_maxim_subphase`

§6.51 – Leader Units

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:417 GH:omdurman-rules/src/lib.rs:417	BritishLeader	<pre>415 /// to claim the Mahdi's Tomb (§9.14). 416 #[derive(Serialize, Deserialize, Clone, Copy, 417 PartialEq, Eq, Hash, Debug, strum::Display)] 418 pub enum BritishLeader { 419 Kitchener,</pre>
omdurman-types/src/lib.rs:932 GH:omdurman-types/src/lib.rs:932	BritishLeader	<pre>930 }, 931 /// Anglo-Egyptian leader (§6.51): movement only. 932 BritishLeader { movement: i32 }, 933 /// Wall-breach marker placed by artillery fire 934 (\$6.63). Not a combat unit. 935 Breech,</pre>
omdurman-types/src/lib.rs:973 GH:omdurman-types/src/lib.rs:973	has_combat_factors	<pre>971 /// British leaders print a movement factor only 972 (\$6.51); other kinds carry 973 /// fire and/or melee factors. Markers carry no 974 stats. 975 pub fn has_combat_factors(self) -> bool { 976 !matches!(977 self,</pre>

Proven by: `omdurman-types::src::lib::unit_kind_fire_and_movement_capability_law_is_exact`

Covered by tests: `omdurman-rules::src::unit_profiles::zero_factor_is_none_not_zero`
`omdurman-rules::src::unit_profiles::kitchener_block_resolves_leaders_friendlys_camel_and_sudanese`
`omdurman-rules::src::unit_profiles::dervish_leader_sections_resolve_leader_and_retinue_per_cell`
`omdurman-rules::src::effects::tests::deploy_rejects_enemy_cohabitation_during_setup`
`omdurman-rules::src::effects::tests::dervish_move_through_lone_ae_leader_hex_eliminate_the_leader`
`omdurman-rules::src::effects::tests::dervish_move_onto_lone_ae_leader_hex_eliminate_the_leader`
`omdurman-rules::src::effects::tests::ae_leader_with_combat_unit_is_not_overrun_by_dervish_move`
`omdurman-rules::src::effects::tests::ae_leader_eliminated_with_last_combat_unit_in_fire_combat`

§6.52 – Anglo-Egyptian Friendlys Brigade

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:629 GH:omdurman-rules/src/lib.rs:629	is_friendlys	<pre>627 /// "Friendlys" units obey several special rules 628 (\$5.21, \$5.23, \$6.52, 629 /// \$9.14 victory conditions). 630 pub fn is_friendlys(&self) -> bool { 631 matches!(632 self,</pre>
omdurman-types/src/lib.rs:1015 GH:omdurman-types/src/lib.rs:1015	Friendlys	<pre>1013 /// Native volunteer brigade -- the Shaggyeh 1014 (\$6.52). Do not receive 1015 /// brigade integrity (\$5.54 enumerates only 1016 British/Egyptian/Sudanese). 1017 Friendlys,</pre>

		1016 } 1017
--	--	------------------

Proven by: `omdurman-rules::src::effects::fire::range_table_player_for_routes_friendlys_and_fok_to_the_dervish_table`

Covered by tests: `omdurman-rules::src::unit_profiles::friendlys_counters_score_by_bank_not_as_leaders`
`omdurman-rules::src::effects::tests::friendlys_validate_and_resolve_on_dervish_table`

§6.53 – Royal Engineers demolition

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:600 GH:omdurman-rules/src/lib.rs:600	RoyalEngineers	598 /// The Royal Engineers (§6.53) -- a *specific* unit, not a class, so we 599 /// model it explicitly. 600 RoyalEngineers, 601 } 602
omdurman-rules/src/lib.rs:835 GH:omdurman-rules/src/lib.rs:835	demolishing	833 /// Set when the Royal Engineers are committed to a demolition this turn 834 /// (§6.53) -- neither offensive fire nor melee allowed that turn. 835 pub demolishing: bool, 836 /// Set when a gunboat has lost its engines to a river mine (§10.12, roll 837 /// 5-7): it may no longer move under power and instead drifts two hexes per
omdurman-rules/src/effects/effect.rs:175 GH:omdurman-rules/src/effects/effect.rs:175	Demolition	173 174 /// Royal Engineers demolition (rulebook §6.53). 175 Demolition { 176 unit_id: UnitId, 177 target: DemolitionTarget,
omdurman-rules/src/effects/setup.rs:35 GH:omdurman-rules/src/effects/setup.rs:35	apply_demolition	33 /// resolution happens at end of turn via [`apply_resolve_demolition`], which 34 /// checks the engineer is still adjacent and undisrupted. 35 pub fn apply_demolition(36 state: &mut GameState, 37 unit_id: UnitId,
omdurman-rules/src/lib.rs:1061 GH:omdurman-rules/src/lib.rs:1061	DemolitionTarget	1059 /// disrupted or driven off. 1060 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1061 pub enum DemolitionTarget { 1062 Fort(UnitId), 1063 WallHexside(HexsideRef),
omdurman-rules/src/effects/state.rs:2136 GH:omdurman-rules/src/effects/state.rs:2136	demolition_targets	2134 /// the rules would accept. Empty when the unit doesn't exist or has no 2135 /// adjacent target. 2136 pub fn demolition_targets(&self, unit_id: UnitId) -> Vec<DemolitionTarget> { 2137 let Ok(unit) = self.unit_or_err(unit_id) else { 2138 return Vec::new();

Covered by tests: `omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_disrupted`
`omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_moved_away`
`omdurman-rules::src::effects::tests::demolition_targets_finds_adjacent_fort_and_wall`

§6.54 – Forts

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:868 GH:omdurman-rules/src/lib.rs:868	ZocReason	<pre>866 /// Used by the engine when answering "is this hex in an enemy ZOC?". 867 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 868 pub enum ZocReason { 869 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 870 /// leader (§5.41) projects ZOC into each of its six adjacent hexes.</pre>
omdurman-rules/src/lib.rs:875 GH:omdurman-rules/src/lib.rs:875	Fort	<pre>873 GunboatVsGunboat, 874 /// Forts project ZOC out of, but not into, an empty fort (§5.44, §6.54). 875 Fort, 876 /// Walled-city ZOC: extends out through walls and gates but not in, 877 /// across a breach in both directions (§5.44).</pre>
omdurman-rules/src/lib.rs:823 GH:omdurman-rules/src/lib.rs:823	UnitState	<pre>821 /// rather than one big enum. 822 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, Default)] 823 pub struct UnitState { 824 /// Reference table: "Disrupted units: no ZOC; may not move; may not fire 825 /// offensively or defensively; may not melee; are turned face up at the</pre>
omdurman-rules/src/lib.rs:976 GH:omdurman-rules/src/lib.rs:976	FireAttack	<pre>974 /// modifiers (rulebook §6). 975 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq, Eq)] 976 pub struct FireAttack { 977 pub firing_player: Player, 978 pub phase: Phase,</pre>
omdurman-rules/src/lib.rs:991 GH:omdurman-rules/src/lib.rs:991	FireAttack::net_modifier	<pre>989 impl FireAttack { 990 /// Sum of all fire modifiers applied to this attack (rulebook §6.24). 991 pub fn net_modifier(&self) -> i16 { 992 // Saturating fold rather than `sum()`: the modifier list is unbounded 993 // and arrives over the network, so a plain sum can overflow `i16`.</pre>

Covered by tests: [omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_enemy_fort](#)

§6.61 – Only artillery may fire at gunboats; 3+ to sink

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Only artillery may fire at gunboats. A result of 3 or more on the combat results table is required to sink a gunboat. Any other result is a miss.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:476 GH:omdurman-rules/src/lib.rs:476	WeaponClass	<pre>474 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display, 475)] 476 pub enum WeaponClass { 477 /// Dervish spears and swords -- no ranged fire at all. 478 Melee,</pre>

Covered by tests: `omdurman-rules::src::effects::tests::rifles_may_not_sink_a_gunboat`
`omdurman-rules::src::effects::tests::artillery_sinks_gunboat_only_on_three_plus`

§6.62 – Only artillery may fire at forts; 2+ to destroy

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Only artillery may fire at forts. A result of 2 or more on the combat results table is required to eliminate a fort. Any other result is a miss. If the fort contains any enemy units at the instant it is destroyed, one unit is eliminated with the fort.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:476 GH:omdurman-rules/src/lib.rs:476	WeaponClass	<pre>474 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, 475 PartialOrd, Ord, Hash, Debug, strum::Display, 476 pub enum WeaponClass { 477 /// Dervish spears and swords -- no ranged fire at 478 all. 479 Melee,</pre>

Covered by tests: `omdurman-rules::src::effects::tests::artillery_destroys_fort_on_two_or_better_only`

§6.63 – Only artillery may breach wall hexsides; 2+ required

implemented

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Only artillery may fire to breach a wall hexside of Khartoum or the walled city of Omdurman. A result of 2 or more on the combat results table is required to breach a wall. Any other result is a miss. The effect of the breach is to negate the wall hexside for line of sight purposes. Place a “BREACH” marker in an adjacent hex so that the arrow points to the breached hexside. If any enemy units are adjacent to the wall hexside at the instant it is breached, one enemy unit is eliminated.

****6.64) Howitzer fire:**** Five units in the game have howitzer fire capability. These are the five named British gunboats. They may fire their artillery factor as direct fire during the Direct Fire Subphase (see 4 and 6.41) and may then fire the same artillery factor as howitzer fire during the Maxim Second Fire and Howitzer Subphase (see 4 and 6.42). Exception: no howitzer fire is allowed during night game turns. To fire howitzer fire, select any target hex between 4 and 10 hexes from the firing gunboat (ignoring the Line of Sight Table) and roll the ten-sided die twice. The first die roll is the Combat Results Table die roll and the second roll is the impact hex die roll. Refer to the Howitzer Fire Scattergram on the marchsheet for the impact hex. The designated target hex is hit on a roll of 7-10. Once a howitzer
(see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:198 GH:omdurman-types/src/lib.rs:198	Breach	<pre>196 /// Breach in a wall (artillery/§6.63 or Royal 197 Engineers/§6.53). ZOC both 198 /// ways; LOS no longer blocked across the hexside. 199 Breach, 200 /// Khor -- gully/wadi. ZOCs do not extend across 201 (\$5.44); advance after 202 /// combat may not cross (§6.82).</pre>
omdurman-rules/src/effects/effect.rs:278 GH:omdurman-rules/src/effects/effect.rs:278	ArtilleryBreachWall	<pre>276 /// pre-rolled d10 used for the CRT lookup; range/ 277 LOS are re-derived by the 278 /// engine from the firers and `target`. 279 ArtilleryBreachWall { 280 firers: Vec<UnitId>, 281 target: HexsideRef,</pre>
omdurman-rules/src/effects/fire.rs:658	apply_artillery_breach_v	<pre>656 /// artillery's CRT roll -- the rulebook specifies the 657 same "2+ required"</pre>

GH:omdurman-rules/src/effects/fire.rs:658		<pre> 657 /// threshold for both trigger styles. 658 pub fn apply_artillery_breach_wall(659 state: &mut GameState, 660 firers: &[UnitId], </pre>
omdurman-rules/src/effects/state.rs:1335 GH:omdurman-rules/src/effects/state.rs:1335	can_fire_at_wall	<pre> 1333 /// range band and resolving the CRT – this method only validates one 1334 /// firer at a time. 1335 pub fn can_fire_at_wall(1336 &self, 1337 firer: UnitId, </pre>

Covered by tests: `omdurman-rules::src::unit_profiles::breach_marker_cell_returns_none`
`omdurman-rules::src::effects::tests::artillery_breaches_wall_only_on_crt_two_or_better`
`omdurman-rules::src::effects::tests::wall_breach_eliminate_one_adjacent_enemy_unit`
`omdurman-rules::src::effects::tests::breach_is_an_opening_for_los_and_effective_kinds`

§6.64 – Howitzer fire

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/howitzer_scatter.rs:8 GH:omdurman-rules/src/howitzer_scatter.rs:8	ScatterHexDirection	<pre> 6 /// the diagram away from the firing player. 7 #[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 8 pub enum ScatterHexDirection { 9 UpperLeft, 10 UpperRight, </pre>
omdurman-rules/src/howitzer_scatter.rs:38 GH:omdurman-rules/src/howitzer_scatter.rs:38	howitzer_scatter	<pre> 36 /// onto a hex-grid offset oriented away from the firer (see 37 /// `GameState::howitzer_impact_hex`). 38 pub fn howitzer_scatter(impact_roll: DieRoll) -> ScatterHexDirection { 39 create::tables_data::SCATTERGRAM[(impact_roll.value() - 1) as usize] 40 } </pre>
omdurman-rules/src/lib.rs:428 GH:omdurman-rules/src/lib.rs:428	GunboatId	<pre> 426 /// fire; "old" gunboats do not (rulebook §2.32). 427 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 428 pub enum GunboatId { 429 /// One of the five new-type named gunboats with howitzer capability. 430 Named(NamedGunboat), </pre>
omdurman-rules/src/effects/effect.rs:88 GH:omdurman-rules/src/effects/effect.rs:88	HowitzerFire	<pre> 86 /// - CRT result applied to units at impact hex (not the original target). 87 /// - Firers marked as fired. 88 HowitzerFire { 89 attack: FireAttack, 90 combat_results_table_roll: DieRoll, </pre>
omdurman-rules/src/effects/fire.rs:13 GH:omdurman-rules/src/effects/fire.rs:13	apply_howitzer_fire	<pre> 11 12 /// Validate and apply a howitzer fire attack (scatter path) (rulebook §6.64). 13 pub fn apply_howitzer_fire(14 state: &mut GameState, 15 attack: &FireAttack, </pre>
omdurman-rules/src/effects/state.rs:1149 GH:omdurman-rules/src/effects/state.rs:1149	can_fire_at	<pre> 1147 /// modifier in the [`FireAttack`] and is responsible for the LOS gate. 1148 /// (Howitzer fire ignores LOS entirely -- §6.64.) 1149 pub fn can_fire_at(1150 &self, 1151 firer: UnitId, </pre>

Proven

by: `omdurman-rules::src::effects::opposite_is_an_involution` `omdurman-rules::src::effects::step_toward_lands_on_an_adjacent_hex`
`omdurman-rules::src::howitzer_scatter::scatter_is_center_exactly_for_rolls_7_to_10`

Covered

by

tests:

`omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10`
`omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7`
`omdurman-rules::src::howitzer_scatter::howitzer_each_miss_gets_its_printed_hex`
`omdurman-rules::src::unit_profiles::named_and_old_gunboats_resolve`
`omdurman-rules::src::effects::tests::named_gunboat_has_howitzer`
`omdurman-rules::src::effects::tests::named_gunboat_may_fire_howitzer_in_second_subphase`
`omdurman-rules::src::effects::tests::named_gunboat_direct_fire_uses_artillery_weapon`
`omdurman-rules::src::effects::tests::named_gunboat_no_howitzer_at_night`
`omdurman-rules::src::effects::tests::dervish_gunboat_lacks_howitzer`

§6.81 – Moving player may fire with all capable units

implicit

manual page unknown

During Offensive Fire phase, the moving player may fire with all of his units capable of firing, up to their maximum range, within the limitations imposed by the rules of combat.

§6.82 – Advance after combat (offensive fire)

implemented

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If an enemy-occupied hex is vacated as a result of offensive fire, friendly units may advance after combat into the vacated hex. To be eligible to advance, the friendly units must have participated in the attack and must have been adjacent to the vacated hex. Note that artillery may not advance, nor may units advance across a wall hexside (except at a gate or breach). Units may never advance after combat across a khor.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:162 GH:omdurman-rules/src/effects/effect.rs:162	AdvanceAfterCombat	<pre>160 /// **Postconditions:** Unit position moved to `to`; `vacated_by_combat` 161 /// entry consumed. 162 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 163 164 // -- Unit state changes -----</pre>
omdurman-rules/src/effects/dispatch.rs:36 GH:omdurman-rules/src/effects/dispatch.rs:36	apply_advance_after_combat	<pre>34 } 35 GameEffect::AdvanceAfterCombat { unit_id, to } => { 36 apply_advance_after_combat(state, *unit_id, *to) 37 } 38 GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id),</pre>
omdurman-types/src/lib.rs:241 GH:omdurman-types/src/lib.rs:241	blocks_advance_after_combat	<pre>239 240 /// Whether advance-after-combat may *not* cross this side (§6.82, §7.6). 241 pub fn blocks_advance_after_combat(self) -> bool { 242 matches!(243 self,</pre>
omdurman-rules/src/effects/state.rs:2044 GH:omdurman-rules/src/effects/state.rs:2044	can_advance_after_combat	<pre>2042 /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 2043 /// Wall/khor hexside restrictions are not enforced (no hexside map data). 2044 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 2045 let unit = self.unit_or_err(unit_id)?;</pre>

		2046 // §6.7: there is no advance after combat as a result of defensive fire.
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Proven by: omdurman-rules::src::effects::advance_phase_is_atomic omdurman-types::src::lib::hexside_blocking_classifiers_are_exact

Covered by tests: omdurman-rules::src::effects::tests::can_advance_after_combat_rejects_wall_hexside
omdurman-rules::src::effects::tests::can_advance_after_combat_rejects_khor_hexside
omdurman-rules::src::effects::tests::advance_requires_combat_vacated_hex
omdurman-rules::src::effects::tests::advance_requires_participation
omdurman-rules::src::effects::tests::rejected_advance_phase_keeps_vacated_windows

8/8 implemented

§7 – Melee Phase

§7 – Melee Phase (chapter)

implemented

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Melee Phase

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:148 GH:omdurman-rules/src/effects/state.rs:148	PendingMelee	<pre>146 /// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5). 147 #[derive(Serialize, Deserialize, Clone, Debug)] 148 pub struct PendingMelee { pub attack: MeleeAttack, 149 pub attacker_roll: DieRoll,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::declared_melee_blocks_phase_advance](#)

§7.1 – Melee strength printed on counter

implemented

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The melee strength of all units is printed on the counter. Note that gunboats have no melee strength. Gunboats may neither melee attack nor be melee attacked.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:110 GH:omdurman-rules/src/lib.rs:110	MeleeFactor	<pre>108 /// Every possible value from the annotated counter set is a named variant. 109 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 110 pub enum MeleeFactor { One = 1, 111 Three = 3,</pre>
omdurman-rules/src/lib.rs:121 GH:omdurman-rules/src/lib.rs:121	MeleeFactor::sum	<pre>119 impl MeleeFactor { 120 /// Sum multiple melee factors (rulebook §7.1). 121 pub fn sum<'a>(factors: impl IntoIterator<Item = &'a MeleeFactor>) -> u16 { 122 factors.into_iter().map(f f.value()).sum() 123 }</pre>
omdurman-types/src/lib.rs:956 GH:omdurman-types/src/lib.rs:956	may_be_melee_attacked	<pre>954 955 /// Gunboats neither attack nor are attacked in melee (§7.1). 956 pub fn may_be_melee_attacked(self) -> bool { 957 !matches!(self, UnitKind::Gunboat { .. }) 958 }</pre>

Proven by: [omdurman-types::src::lib::unit_kind_melee_capability_law_is_exact](#)

Covered by tests: [omdurman-rules::src::lib::melee_factor_values_and_sum](#) [omdurman-rules::src::lib::unit_kind_melee_capability](#)

§7.2 – Melee adjacent only, not across wall hexsides

implemented

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Melee simulates the hand-to-hand fighting of the period. Units may melee attack adjacent enemy units only. Units may not melee attack across a wall hexside, but may melee attack through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:236 GH:omdurman-types/src/lib.rs:236	blocks_melee	<pre>234 /// Whether melee may *not* be made across this side (§7.2). Gates and 235 /// breaches are passable to melee. 236 pub fn blocks_melee(self) -> bool { 237 matches!(self, HexsideKind::Wall HexsideKind::ZaribaThornHedge) 238 }</pre>
omdurman-rules/src/effects/state.rs:1426 GH:omdurman-rules/src/effects/state.rs:1426	can_melee	<pre>1424 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 1425 /// thorn-hedge hexside blocks the attack (§7.2). 1426 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> { 1427 let unit = self.unit_or_err(attacker)?; 1428 }</pre>

Proven by: [omdurman-types::src::lib::hexside_blocking_classifiers_are_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::can_melee_gates_phase_adjacency_and_kind](#)
[omdurman-rules::src::effects::tests::can_melee_rejects_wall_hexside](#)
[omdurman-rules::src::effects::tests::can_melee_rejects_thorn_hedge_hexside](#)
[omdurman-rules::src::effects::tests::can_melee_allows_gate_hexside](#)

§7.3 – Simultaneous melee combat

implemented

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Melee combat is considered simultaneous, so that units eliminated by melee attacks still get a melee combat die roll.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:111 GH:omdurman-rules/src/effects/effect.rs:111	MeleeCombat	<pre>109 /// - Winner may advance into vacated hex (§7.6). 110 /// - Victory points awarded for eliminations. 111 MeleeCombat { 112 attack: MeleeAttack, 113 attacker_roll: DieRoll,</pre>
omdurman-rules/src/effects/melee.rs:4 GH:omdurman-rules/src/effects/melee.rs:4	apply_melee_combat	<pre>2 3 /// Apply a simultaneous melee combat between two adjacent hexes (rulebook §7). 4 pub fn apply_melee_combat(5 state: &mut GameState, 6 attack: &MeleeAttack,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::melee_resolves_simultaneously](#)

§7.4 – Who may melee attack / defend

implemented

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Only infantry, cavalry, camel units, and Dervish leaders may melee attack. All units (except gunboats — see 7.1) may melee defend.

See also: §7.1

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:945 GH:omdurman-types/src/lib.rs:945	may_melee_attack	<pre> 943 /// Rulebook §7.4 -- only infantry, cavalry, camel and Dervish leaders may 944 /// melee *attack*. All others (except gunboats) may melee *defend* (§7.1). 945 pub fn may_melee_attack(self) -> bool { matches!(946 self, </pre>
omdurman-types/src/lib.rs:882 GH:omdurman-types/src/lib.rs:882	UnitKind	<pre> 880 /// `Some(UnitKind::Marker)` or `None`. 881 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 882 pub enum UnitKind { 883 /// Foot infantry (§2.3): fire / melee / movement. 884 Infantry { </pre>
omdurman-types/src/lib.rs:724 GH:omdurman-types/src/lib.rs:724	DervishTribe	<pre> 722 strum::EnumIter, 723)] 724 pub enum DervishTribe { 725 Baggara, 726 Jaalin, </pre>
omdurman-rules/src/effects/state.rs:1426 GH:omdurman-rules/src/effects/state.rs:1426	can_melee	<pre> 1424 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 1425 /// thorn-hedge hexside blocks the attack (§7.2). 1426 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> { 1427 let unit = self.unit_or_err(attacker)?; 1428 </pre>

Proven by: [omdurman-types::src::lib::unit_kind_melee_capability_law_is_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::can_melee_gates_phase_adjacency_and_kind](#)

§7.5 – Cavalry/camel retreat before melee

implemented

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Cavalry and camel units may retreat two hexes from an infantry melee attack. Note, however, that only one retreat per unit per turn is permitted. Thus, if their retreat places them adjacent to enemy units whose melee attacks have not yet been resolved, those enemy units may elect to attack the retreating unit(s).

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:148 GH:omdurman-rules/src/effects/effect.rs:148	RetreatBeforeMelee	<pre> 146 /// 147 /// **Postconditions:** Unit position moved to `to`. 148 RetreatBeforeMelee { unit_id: UnitId, to: HexCoord }, 149 150 /// An attacking unit advances into a hex vacated by combat (rulebook §6.82 </pre>
omdurman-rules/src/effects/dispatch.rs:33 GH:omdurman-rules/src/effects/dispatch.rs:33	apply_retreat_before_melee	<pre> 31 GameEffect::ResolveMelee => apply_resolve_melee(state), 32 GameEffect::RetreatBeforeMelee { unit_id, to } => { 33 apply_retreat_before_melee(state, *unit_id, *to) 34 } 35 GameEffect::AdvanceAfterCombat { unit_id, to } => { </pre>
omdurman-types/src/lib.rs:962	may_retreat_before_melee	<pre> 960 /// Cavalry and camel units may retreat two hexes from an infantry melee 961 /// attack (§7.5). 962 pub fn may_retreat_before_melee(self) -> bool { </pre>

GH:omdurman-types/src/lib.rs:962		<pre> 963 matches!(self, UnitKind::Cavalry { .. } UnitKind::Camel { .. }) 964 }</pre>
omdurman-rules/src/lib.rs:187 GH:omdurman-rules/src/lib.rs:187	HexDistance	<pre> 185 /// (rulebook §6.22, §7.5). 186 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 187 pub struct HexDistance(u16); 188 189 impl HexDistance {</pre>
omdurman-rules/src/effects/state.rs:1975 GH:omdurman-rules/src/effects/state.rs:1975	can_retreat_before_melee	<pre> 1973 /// two hexes away and empty. (Does not verify the attacker is infantry -- 1974 /// the caller offers the retreat only in response to one.) 1975 pub fn can_retreat_before_melee(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 1976 let unit = self.unit_or_err(unit_id)?; 1977 if !matches!(self.phase, Phase::Melee) {</pre>

Proven by: `omdurman-rules::src::effects::resolve_melee_is_atomic` `omdurman-rules::src::effects::advance_phase_is_atomic`
`omdurman-types::src::lib::unit_kind_melee_capability_law_is_exact`

Covered by tests: `omdurman-rules::src::effects::tests::retreat_before_melee_only_cavalry_two_hexes`
`omdurman-rules::src::effects::tests::retreat_opens_window_only_when_hex_empties`
`omdurman-rules::src::effects::tests::rejected_resolve_melee_keeps_the_declaration`

§7.6 – Advance after melee

implemented

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If a melee attack eliminates all of the defenders in an adjacent hex, the Dervish player **MUST** advance into the vacated hex. To be eligible to advance, the Dervish units must have been adjacent to the vacated hex and participated in the melee attack that eliminated the defenders. All surviving eligible Dervish units **MUST** advance, up to the stacking limit. The Anglo-Egyptian player may advance if desired. Note that only attacking units may advance.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:162 GH:omdurman-rules/src/effects/effect.rs:162	AdvanceAfterCombat	<pre> 160 /// **Postconditions:** Unit position moved to `to`, `vacated_by_combat` 161 /// entry consumed. 162 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 163 164 // -- Unit state changes -----</pre>
omdurman-rules/src/effects/dispatch.rs:36 GH:omdurman-rules/src/effects/dispatch.rs:36	apply_advance_after_combat	<pre> 34 } 35 GameEffect::AdvanceAfterCombat { unit_id, to } => { 36 apply_advance_after_combat(state, *unit_id, *to) 37 } 38 GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id),</pre>
omdurman-rules/src/effects/state.rs:2044 GH:omdurman-rules/src/effects/state.rs:2044	can_advance_after_combat	<pre> 2042 /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 2043 /// Wall/khor hexside restrictions are not enforced (no hexside map data). 2044 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 2045 let unit = self.unit_or_err(unit_id)?; 2046 // §6.7: there is no advance after combat as a result of defensive fire.</pre>

Proven by: `omdurman-rules::src::effects::advance_phase_is_atomic`

Covered by tests: `omdurman-rules::src::effects::tests::dervish_advance_after_melee_is_mandatory`
`omdurman-rules::src::effects::tests::rejected_advance_phase_keeps_vacated_windows`
`omdurman-rules::src::effects::tests::melee_mandatory_advance_leaders_are_free_stacking`

§7.7 – Melee modifiers

implemented

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To resolve melee, both the attacker and the defender roll on the Combat Results Table and apply the applicable melee modifier to their die roll. The Dervish player receives a +2 melee modifier, the Anglo-Egyptian player receives a +1 melee modifier. No terrain modifiers are applied to melee combat (Exception: Zariba hexsides in the historical scenario and the campaign game, if constructed — see 9.23). Melee losses must be taken from meleeing units first!

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1019 GH:omdurman-rules/src/lib.rs:1019	MeleeModifier	<pre>1017 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1018 1019 pub enum MeleeModifier { 1020 /// +2 to all Dervish melee rolls (§7.7). 1021 DervishStandard,</pre>
omdurman-rules/src/lib.rs:1042 GH:omdurman-rules/src/lib.rs:1042	MeleeAttack	<pre>1040 /// A melee attack: simultaneous, both sides roll on the Combat Results Table (§7.3, §7.7). 1041 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq, Eq)] 1042 pub struct MeleeAttack { 1043 pub attacker_player: Player, 1044 pub attacker_hex: HexCoord,</pre>
omdurman-rules/src/lib.rs:1023 GH:omdurman-rules/src/lib.rs:1023	MeleeModifier::Anglo-EgyptianStandard	<pre>1021 DervishStandard, 1022 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 1023 AngloEgyptianStandard, 1024 /// Inverted to -2 when Dervish units melee-attack across a trench into 1025 /// an entrenched defender (§9.232).</pre>
omdurman-rules/src/lib.rs:1026 GH:omdurman-rules/src/lib.rs:1026	MeleeModifier::DervishVsDefender	<pre>1024 /// Inverted to -2 when Dervish units melee-attack across a trench into 1025 /// an entrenched defender (§9.232). 1026 DervishVsTrenchedDefender, 1027 } 1028 </pre>
omdurman-rules/src/lib.rs:1021 GH:omdurman-rules/src/lib.rs:1021	MeleeModifier::DervishStandard	<pre>1019 pub enum MeleeModifier { 1020 /// +2 to all Dervish melee rolls (§7.7). 1021 DervishStandard, 1022 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 1023 AngloEgyptianStandard,</pre>

Proven by: `omdurman-rules::src::lib::melee_modifier_keeps_roll_legal`

Covered by tests: `omdurman-rules::src::effects::tests::melee_resolves_simultaneously`
`omdurman-rules::src::effects::tests::melee_modifiers_are_engine_derived_and_mismatches_rejected`

2/3 implemented

§8 – Night Game Turns

§8 – Night Game Turns

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Night Game Turns

§8.1 – Night effects

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The effects of night game turns are: a) all Anglo-Egyptian movement allowances are halved (round down), b) there is no Anglo-Egyptian howitzer fire, and c) all fire ranges are halved for both sides (round down, but range 1 stays range 1). Range effects on fire combat are the same as during day game turns. For example, an Anglo-Egyptian infantry unit firing at night will be doubled at range 1, normal at range 2, and may not fire at range 3 or greater.

****8.2) Dervish Desertion Roll:**** Once each campaign game, during the first night turn of the game, the Dervish player rolls one die to see how many of his units desert. The roll is made during the movement phase and the number of deserting Dervish units is equal to 1½ times the roll of one die. The Dervish player may choose which units desert by merely removing them from the mapsheet. The KHALIFA unit, gunboats, artillery units, and forts are the only Dervish units that may not be chosen. No victory points are awarded to the Anglo-Egyptian player for deserting Dervishes.

(see manual for full text)

See also: §8.2

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:156 GH:omdurman-rules/src/lib.rs:156	MovementAllowance::halve	<pre>154 impl MovementAllowance { 155 /// Night movement allowance = halved (round down) (rulebook §8.1, §5.11). 156 pub fn halve(self) -> Self { 157 let v = self.value() / 2; 158 MovementAllowance::try_from(v).expect("halved value always a named variant")</pre>
omdurman-types/src/lib.rs:789 GH:omdurman-types/src/lib.rs:789	DayNight	<pre>787 /// (rulebook §8.1). 788 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 789 pub enum DayNight { 790 Day, 791 Night,</pre>
omdurman-rules/src/range_effects.rs:71 GH:omdurman-rules/src/range_effects.rs:71	night_max_range	<pre>69 70 /// The halved maximum range at night (§8.1): round down, minimum 1. 71 pub fn night_max_range(weapon: WeaponClass, ae: bool) - > u8 { 72 let day = max_day_range(weapon, ae); 73 if day <= 1 { 1 } else { day / 2 }</pre>
omdurman-rules/src/lib.rs:1514 GH:omdurman-rules/src/lib.rs:1514	effective_movement_at_night	<pre>1512 /// Apply night-turn movement halving for Anglo- Egyptian units (§8.1): all 1513 /// Anglo-Egyptian movement allowances are halved (round down). 1514 pub fn effective_movement_at_night(1515 allowance: MovementAllowance, 1516 player: Player,</pre>

Proven by: `omdurman-rules::src::lib::movement_allowance_half_never_panics`
`omdurman-rules::src::lib::night_halving_is_ae_only_and_day_neutral`
`omdurman-rules::src::range_effects::night_cap_is_halved_day_max_floored_at_one`
`omdurman-rules::src::range_effects::night_range_effects_gates_the_day_band_by_the_cap`
`omdurman-rules::src::effects::fire::night_capped_distance_is_some_exactly_within_the_night_max`

Covered by tests: `omdurman-rules::src::range_effects::night_max_ranges` `omdurman-rules::src::range_effects::night_max_ranges_remaining`
`omdurman-rules::src::range_effects::ae_rifle_at_night_matches_rulebook_example`
`omdurman-rules::src::range_effects::max_day_range_all_combos`
`omdurman-rules::src::effects::tests::night_movement_overlay_allowance_halved`

§8.2 – Dervish Desertion Roll

implemented

manual page unknown

File	Symbol	Code Snippet
<code>omdurman-rules/src/effects/effect.rs:190</code> GH:omdurman-rules/src/effects/effect.rs:190	DervishDesertion	<pre>188 /// the effect. The Khalifa, gunboats, artillery, 189 and forts may not be 190 chosen. 191 DervishDesertion { 192 roll: DieRoll, 193 deserters: Vec<UnitId>,</pre>
<code>omdurman-rules/src/turn_track.rs:63</code> GH:omdurman-rules/src/turn_track.rs:63	DervishDesertion	<pre>61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the 63 first night turn. 64 DervishDesertion, 65 /// Dervish reinforcements are available. 66 DervishReinforcements,</pre>
<code>omdurman-rules/src/turn_track.rs:60</code> GH:omdurman-rules/src/turn_track.rs:60	TurnEvent	<pre>58 /// Special events that occur on specific turns 59 (rulebook §8.2, §9.112, §9.113). 60 #[derive(Serialize, Deserialize, Clone, Copy, 61 PartialEq, Eq, Debug)] 62 pub enum TurnEvent { 63 None, 64 /// Dervish desertion roll (§8.2) -- occurs on the 65 first night turn.</pre>

Proven by: `omdurman-rules::src::effects::desertion_count_is_one_and_a_half_times_the_roll`
`omdurman-rules::src::turn_track::desertion_event_is_unique_to_the_first_night_turn`

Covered by tests: `omdurman-rules::src::effects::tests::desertion_count_is_floor_one_and_a_half`
`omdurman-rules::src::turn_track::desertion_on_first_night`
`omdurman-rules::src::effects::tests::desertion_roll_required_before_first_night_movement_ends`

23/33 implemented

§9 – The Scenarios

§9 – The Scenarios

descriptive

manual page unknown

The Scenarios

§9.1 – The Campaign Game

descriptive

manual page unknown

The Campaign Game

§9.2 – The Historical Scenario

descriptive

manual page unknown

The Historical Scenario

Players should note that the historical scenario is an exercise in futility for the Dervish player. It is, however, an interesting demonstration of the absolute imbecility of the Khalifa's generalship and vividly shows the superiority of entrenched firepower over traditional tribal arms in the colonial period.

§9.3 – Bonus Game: Fall of Khartoum

descriptive

manual page unknown

Bonus Game: FALL OF KHARTOUM

§9.11 – Set Up (Campaign)

descriptive

manual page unknown

Set Up

§9.12 – Scenario Length (Campaign)

implemented

manual page unknown

Scenario Length

6:00 am, Sept. 1 through 8:00 am, Sept. 3, 22 Game Turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:11 GH:omdurman-	GameTime	<pre>9 /// starts at one of these twelve times. 10 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 11 pub enum GameTime {</pre>

rules/src/ turn_track.rs:11		12 SixAM, 13 EightAM,
omdurman- rules/src/ turn_track.rs:47 GH:omdurman- rules/src/ turn_track.rs:47	TurnEntry	45 /// A single entry on the Turn Record Track (rulebook \$9.12, §9.22). 46 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 47 pub struct TurnEntry { 48 /// 1-based turn number. 49 pub turn: u8,
omdurman- rules/src/ turn_track.rs:95 GH:omdurman- rules/src/ turn_track.rs:95	CAMPAIGN_TURN_TRACK	93 /// is turn 9, which carries the once-per-game Dervish Desertion Roll (§8.2) -- 94 /// the printed track prints "Dervish Desertion Roll / NIGHT" on that cell. 95 pub const CAMPAIGN_TURN_TRACK: [TurnEntry; 22] = [96 // Row 1, left->right: Sept 1, 6 am -> 8 pm, then the first NIGHT. 97 entry(1, SixAM, Day, TurnEvent::None),
omdurman- rules/src/turn_track.rs:242 GH:omdurman- rules/src/ turn_track.rs:242	TurnLabel	240 /// printed cell; `Blank` is for unused positions in the 9×3 grid. 241 #[derive(Serialize, Deserialize, Debug, Clone, Copy, PartialEq, Eq)] 242 pub enum TurnLabel { 243 Blank, 244 /// "SEPT. 1\n6:00 am" -- day header plus time (turn 1).

Proven by: omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario
omdurman-rules::src::turn_track::turn_labels_agree_with_the_rule_bearing_track

Covered
by tests: omdurman-rules::src::turn_track::campaign_track_22_turns omdurman-rules::src::turn_track::desertion_on_first_night
omdurman-rules::src::turn_track::campaign_track_label_and_day_night_agree
omdurman-rules::src::turn_track::game_time_display_all_variants omdurman-rules::src::turn_track::turn_label_display
omdurman-rules::src::turn_track::turn_label_out_of_range_is_none
omdurman-rules::src::effects::tests::game_over_after_campaign_turns

§9.13 – Special Rules (Campaign)

descriptive
manual page unknown

- Special Rules
- None.

§9.14 – Victory Conditions (Campaign)

implemented
manual page unknown

Victory Conditions

The Mahdi’s Tomb in Omdurman was not only the tallest structure in the entire Sudan in 1898, it was also a Dervish holy shrine. Its loss or destruction would be a severe blow to the Mahdist cause. It is accordingly assigned 25 victory points which are awarded to the player who controls it at the conclusion of play. The Dervish player controls it at the start of play. As a tactical note, the Anglo-Egyptian player will find a decisive victory almost impossible unless he takes the Mahdi’s Tomb from the Dervish player. To take the Tomb hex, it must be occupied by one British leader plus any one non-“Friendlies” Anglo-Egyptian combat unit (both undisrupted) at the conclusion of play.

Additional victory points are awarded as follows:

****Dervish Player receives:****

- 10 pts: each British leader eliminated

(see manual for full text)

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1168 GH:omdurman-rules/src/lib.rs:1168	VpSource	<pre>1166 /// the manual and the engine. 1167 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1168 pub enum VpSource { 1169 // ----- Anglo-Egyptian player receives: 1170 /// Mahdi's Tomb control at conclusion of play (\$9.14).</pre>
omdurman-rules/src/lib.rs:1194 GH:omdurman-rules/src/lib.rs:1194	VpSource::points	<pre>1192 impl VpSource { 1193 /// VP awarded to `who_scores()` (rulebook §9.14). 1194 pub fn points(self) -> VictoryPoints { 1195 match self { 1196 VpSource::MahdisTomb => VictoryPoints::new(25),</pre>
omdurman-rules/src/lib.rs:1209 GH:omdurman-rules/src/lib.rs:1209	VpSource::who_scores	<pre>1207 1208 /// Which player receives these victory points (rulebook §9.14). 1209 pub fn who_scores(self) -> Player { 1210 match self { 1211 VpSource::MahdisTomb</pre>
omdurman-rules/src/lib.rs:1244 GH:omdurman-rules/src/lib.rs:1244	VictoryLedger	<pre>1242 /// Cumulative victory ledger for one scenario (rulebook §9.14). 1243 #[derive(Serialize, Deserialize, Clone, Debug, Default)] 1244 pub struct VictoryLedger { 1245 pub events: Vec<VpEvent>, 1246 }</pre>
omdurman-rules/src/lib.rs:1250 GH:omdurman-rules/src/lib.rs:1250	VpEvent	<pre>1248 /// A single victory-point scoring event (rulebook §9.14). 1249 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1250 pub struct VpEvent { 1251 pub turn: GameTurnIndex, 1252 pub source: VpSource,</pre>
omdurman-rules/src/lib.rs:1257 GH:omdurman-rules/src/lib.rs:1257	VictoryLedger::total_for	<pre>1255 impl VictoryLedger { 1256 /// Total victory points earned by a given player (rulebook §9.14). 1257 pub fn total_for(&self, player: Player) -> VictoryPoints { 1258 VictoryPoints(1259 self.events</pre>
omdurman-rules/src/lib.rs:1269 GH:omdurman-rules/src/lib.rs:1269	VictoryLedger::superiority	<pre>1267 /// Net superiority: positive = Anglo-Egyptian ahead, negative = Dervish ahead 1268 /// (rulebook §9.14). 1269 pub fn superiority(&self) -> VictoryPoints { 1270 VictoryPoints(1271 self.total_for(Player::AngloEgyptian).value() - self.total_for(Player::Dervish).value()),</pre>
omdurman-rules/src/lib.rs:1289 GH:omdurman-rules/src/lib.rs:1289	CampaignVictoryLevel	<pre>1287 /// Campaign-game victory levels (§9.14). 1288 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1289 pub enum CampaignVictoryLevel { 1290 Draw, 1291 Marginal(Player),</pre>
omdurman-rules/src/lib.rs:1298 GH:omdurman-rules/src/lib.rs:1298	CampaignVictoryLevel::from_superiority	<pre>1296 impl CampaignVictoryLevel { 1297 /// Assign a level from the net superiority (\$9.14). 1298 pub fn from_superiority(s: VictoryPoints) -> Self { 1299 let net = s.0;</pre>

		1300 // Positive -> Anglo-Egyptian thresholds: 15/30/50
omdurman-rules/src/effects/movement.rs:194 GH:omdurman-rules/src/effects/movement.rs:194	score_elimination	192 .find_unit(leader) 193 .is_some_and(u u.profile.identity.is_gordon()); 194 score_elimination(state, leader, ElimCause::Combat); 195 state.units.retain(u u.id != leader); 196 if is_gordon {
omdurman-rules/src/lib.rs:239 GH:omdurman-rules/src/lib.rs:239	VictoryPoints	237 /// (rulebook §9.14). 238 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug, Default)] 239 pub struct VictoryPoints(i32); 240 241 impl VictoryPoints {

Proven by:

omdurman-rules::src::lib::campaign_victory_levels_match_manual_superiority_table
omdurman-rules::src::lib::vp_source_points_and_scorer_match_the_printed_schedule
omdurman-rules::src::effects::victory::vp_source_for_routes_every_elimination_to_the_printed_source
omdurman-rules::src::effects::victory::score_elimination_records_exactly_what_it_scores

Covered by tests:

omdurman-rules::src::effects::tests::friendlies_bank_scores_by_side
omdurman-rules::src::effects::tests::mahdis_tomb_not_scored_without_a_leader
omdurman-rules::src::effects::tests::mahdis_tomb_scores_for_anglo_egyptian_when_held
omdurman-rules::src::lib::vp_source_attributes

§9.21 – Set Up (Historical)

descriptive

manual page unknown

Set Up

§9.22 – Scenario Length (Historical)

implemented

manual page unknown

Scenario Length

6:00 am, September 2 through 12:00 noon, September 2. Four game turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:132 GH:omdurman-rules/src/turn_track.rs:132	HISTORICAL_TURN	130 131 /// Historical scenario track (§9.22 -- 4 turns, Sept 2 6:00 am -> 12:00 pm). 132 pub const HISTORICAL_TURN_TRACK: [TurnEntry; 4] = [133 TurnEntry { 134 turn: 1,

Proven by:

omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario

Covered by tests:

omdurman-rules::src::turn_track::historical_turn_all_four_turns

§9.23 – Special Rule: The Zariba

descriptive

manual page unknown

Special Rule: “The Zariba”

****9.231)** Thorn hedge hexsides:****** -2 to die roll on all Dervish fire attacks; may not melee across in either direction; may not advance after combat across in either direction.

****9.232)** Trench hexsides:****** -4 to die roll on all Dervish fire attacks vs. entrenched units only; -2 (instead of +2) melee modifier to Dervish units melee attacking an entrenched unit; entrenched units may be fired “over” in both directions (i.e. they do not block line of sight); units are considered “entrenched” if they are directly adjacent to (and on the Nile River side of) a trench hexside.

(see manual for full text)

See also: §9.231, §9.232

§9.24 – Victory Conditions (Historical)

implemented
manual page unknown

Victory Conditions

Victory Levels are based solely on eliminating enemy units while conserving your own force as much as possible.

| Victory Level | Anglo-Egyptian Player (Dervish units eliminated) | Dervish Player (Anglo-Egyptian units eliminated)
| |—|—|—| | 5 — DECISIVE | 100+ | 30+ | | 4 — STRATEGIC | 60–99 | 15–29 | | 3 — TACTICAL | 45–59 | 10–14 | | 2 — MARGINAL | 30–44 | 5–9 | | 1 — DRAW | 0–29 | 0–4 |

The lower value victory level is then subtracted from the higher level to determine a player’s net victory. For example, if the Anglo-Egyptian player eliminates 104 Dervish units (decisive victory) but loses 18 units doing it (Dervish Strategic), the Anglo-Egyptian player only nets out with a draw (decisive worth 5 minus strategic worth 4 = 1, draw).

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1328 GH:omdurman-rules/src/lib.rs:1328	HistoricalVictoryLevel	<pre>1326 /// draw"). 1327 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)] 1328 pub enum HistoricalVictoryLevel { 1329 Draw = 1, 1330 Marginal = 2,</pre>

Proven by: [omdurman-rules::src::lib::historical_victory_ladders_match_manual_bands](#)

Covered by tests:
[omdurman-rules::src::lib::historical_victory_level_for_dervish](#) [omdurman-rules::src::lib::historical_victory_level_for_anglo_egyptian](#)

§9.31 – Bonus game map

implemented
manual page unknown

Only the small FALL OF KHARTOUM scenario map is used for this game.

File	Symbol	Code Snippet
omdurman-rules/src/board_data.rs:37 GH:omdurman-rules/src/board_data.rs:37	fall_of_khartoum_map	<pre>35 36 /// The Fall-of-Khartoum board (§9.3). 37 pub fn fall_of_khartoum_map_data() -> MapData { 38 FOK.get_or_init(parse(FOK_RON)).clone() 39 }</pre>

Covered by tests: `omdurman-app::src::tests::scenario_maps_to_board` `omdurman-app::src::tests::start_game_scenario_selects_board`

§9.32 – Set Up (Bonus)

`descriptive`
manual page unknown

Set Up

§9.33 – Scenario Length (Bonus)

`implemented`
manual page unknown

Scenario Length
Variable, see victory conditions (9.35). Rarely lasts five turns.

See also: §9.35

File	Symbol	Code Snippet
<code>omdurman-rules/src/turn_track.rs:172</code> <code>GH:omdurman-rules/src/turn_track.rs:172</code>	<code>FALL_OF_KHARTOUM</code>	<pre>170 /// (the rulebook fixes none); only `day_night` is 171 rule-bearing (night halves 172 /// Anglo-Egyptian movement and ranges and bars 173 howitzer fire, §8.1). 174 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [175 TurnEntry { 176 turn: 1,</pre>

Proven by: `omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario`
`omdurman-rules::src::turn_track::fall_of_khartoum_night_window_is_turns_one_and_two`

Covered by tests: `omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night`
`omdurman-rules::src::turn_track::fall_of_khartoum_turns_3_to_8_are_day`

§9.34 – Special Rules (Bonus)

`descriptive`
manual page unknown

Special Rules

§9.35 – Victory Conditions (Bonus)

`implemented`
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Victory Conditions
Victory is determined by how many turns it takes the Dervish player to eliminate the GORDON leader unit and how many Dervish units are eliminated:

- Dervish decisive: eliminate GORDON turn four or sooner.
- Dervish tactical: eliminate GORDON turn five.
- Dervish marginal: eliminate GORDON turn six.
- British marginal: GORDON survives end of turn six.
- British tactical: GORDON survives end of turn seven.
- British decisive: GORDON survives end of turn eight.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1446 GH:omdurman-rules/src/lib.rs:1446	FoKVictoryLevel::resolve	<pre> 1444 /// rulebook: GORDON dies turn 5 (tactical) with 1445 24 Dervish losses (-2 1446 /// levels) nets a British marginal. 1447 pub fn resolve(gordon_died_turn: Option<u8>, 1448 scenario_end_turn: u8, dervish_lost: i16) -> Self { 1449 let base = Self::base(gordon_died_turn, 1450 scenario_end_turn); 1451 let base_idx = Self::LADDER </pre>

Proven by: [omdurman-rules::src::lib::fok_victory_ladder_penalties_shift_monotonically](#)

Covered

by tests: [omdurman-rules::src::lib::fok_victory_level_worked_example](#) [omdurman-rules::src::lib::fok_victory_level_gordon_died_early](#) [omdurman-rules::src::lib::fok_victory_level_late_gordon_death](#) [omdurman-rules::src::lib::fok_victory_level_gordon_survived](#)

§9.111 – Dervish set up (Campaign)

implemented

manual page unknown

Dervish player sets up first, moves second.

- Isa Zachneih infantry unit: anywhere on the east bank, in or south of El Debeba.
- KHALIFA ABDULLAH: in the walled city of Omdurman, in either palace hex.
- 3 artillery units, and all Taiasha units: anywhere in the walled city of Omdurman.
- 17 forts: anywhere on the mapsheet south of the Khor Shambat on the west bank, and/or south of all Halfaya hut hexes on the east bank and Nile River islands.
- 2 gunboats: any south edge Nile River hexes.

****9.112) Dervish reinforcements:**** all reinforcements enter on the west edge of the mapsheet, south of the Khor Shambat. Each unit pays the terrain cost of the hex through which it enters, no matter how many units enter through

(see manual for full text)

See also: §9.112

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:358 GH:omdurman-rules/src/effects/state.rs:358	setup_complete	<pre> 356 /// currently shares the same "both sides deployed" 357 gate; when a scenario 358 /// needs a different minimum, branch on 359 `self.scenario` here. 360 pub fn setup_complete(&self) -> Result<(), 361 RuleError> { 362 let has = player { 363 self.units </pre>

Covered

by

tests:

[omdurman-app::src::scenario_setup::campaign_has_no_fixed_placements](#) [omdurman-rules::src::unit_profiles::hadendowa_first_cell_is_isa_zachneih](#) [omdurman-rules::src::effects::tests::campaign_deployment_is_boat_land_exclusive](#) [omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force](#)

§9.112 – Dervish reinforcements (Campaign)

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:182 GH:omdurman-	PlaceReinforcements	<pre> 180 // -- Reinforcement / placement 181 ----- 182 /// Place reinforcements onto the map (rulebook </pre>

rules/src/effects/ effect.rs:182		<pre> \$9.112, \$9.113). 182 PlaceReinforcements(Vec<UnitPlacement>), 183 184 // -- Scenario-specific ----- </pre>
omdurman- rules/src/effects/ dispatch.rs:44 GH:omdurman- rules/src/effects/ dispatch.rs:44	apply_place_reinforceme	<pre> 42 GameEffect::Demolition { unit_id, target } => apply_demolition(state, *unit_id, *target), 43 GameEffect::PlaceReinforcements(placements) => { 44 apply_place_reinforcements(state, placements) 45 } 46 GameEffect::DervishDesertion { roll, deserters } => { </pre>
omdurman- rules/src/reinforcements.rs:69 GH:omdurman- rules/src/reinforcements.rs:69	dervish_campaign_sched	<pre> 67 /// All reinforcements enter on the west edge, south of the Khor Shambat. 68 /// Each unit pays terrain cost of the hex it enters through. 69 pub fn dervish_campaign_schedule() -> ReinforcementSchedule { 70 ReinforcementSchedule { 71 player: Player::Dervish, </pre>
omdurman- types/src/ lib.rs:524 GH:omdurman- types/src/lib.rs:524	Location	<pre> 522 /// Named map landmarks (rulebook mapsheet, \$9.111, \$9.113, \$9.212 scenarios). 523 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 524 pub enum Location { 525 FortMakran, 526 NorthFort, </pre>
omdurman- types/src/ lib.rs:604 GH:omdurman- types/src/lib.rs:604	SetupLetter	<pre> 602 /// Each letter marks a specific hex where a Dervish leader is placed. 603 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 604 pub enum SetupLetter { 605 Y, 606 K, </pre>
omdurman- types/src/ lib.rs:747 GH:omdurman- types/src/lib.rs:747	Faction	<pre> 745 /// `Some(BrigadeId::friendlies())`. 746 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 747 pub enum Faction { 748 Dervish { 749 tribe: DervishTribe, </pre>

Proven by: [omdurman-rules::src::reinforcements::wave_for_turn_answers_exactly_on_the_printed_schedule](#)

Covered by tests: [omdurman-rules::src::reinforcements::dervish_schedule_has_three_waves](#)
[omdurman-rules::src::reinforcements::dervish_wave_one_has_baggaara_and_three_leaders](#)
[omdurman-rules::src::reinforcements::dervish_wave_two_has_hadendowa](#)
[omdurman-rules::src::reinforcements::dervish_wave_three_is_all_remaining](#)
[omdurman-rules::src::reinforcements::wave_for_turn_returns_correct_wave](#)
[omdurman-rules::src::effects::tests::campaign_reinforcements_gate_by_wave](#)
[omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force](#)

\$9.113 – Anglo-Egyptian reinforcements (Campaign)

implemented

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The Anglo-Egyptian player moves first. There are no Anglo-Egyptian units on the mapsheet at start. The GORDON unit is not used in this scenario.

- The leader units KITCHENER, GATACRE, and HUNTER may enter anytime during the first four game turns and do not count against the 12 units per turn limit. All three leaders must be in play by the end of turn four!

- All gunboats enter through any north edge Nile River hex, paying one movement point for the first hex entered. The “Friendlies” brigade enters through the Abu Alim hut hex on the east bank, paying eight movement points per unit. All other Anglo-Egyptian units enter through the west bank “ANGLO-EGYPTIAN ENTRANCE AREA”, each unit paying one movement point to enter the mapsheet.
- Turn 1) Any three gunboats; “Friendlies” brigade; Egyptian Cavalry; Horse Artillery; and two infantry brigades

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/reinforcements.rs:126 GH:omdurman-rules/src/reinforcements.rs:126	anglo_egyptian_campaign	<pre> 124 /// - "Friendlies" enter via Abu Alim hut on the east bank (8 MP per unit). 125 /// - All other AE units enter via the Anglo-Egyptian Entrance Area (1 MP). 126 pub fn anglo_egyptian_campaign_schedule() -> ReinforcementSchedule { 127 let free_leaders = vec![128 CampaignLeader::British(BritishLeader::Kitchener), </pre>

Proven by: [omdurman-rules::src::reinforcements::wave_for_turn_answers_exactly_on_the_printed_schedule](#)

Covered by tests:

- [omdurman-rules::src::reinforcements::anglo_egyptian_schedule_has_four_waves](#)
- [omdurman-rules::src::reinforcements::anglo_egyptian_leaders_available_each_wave](#)
- [omdurman-rules::src::reinforcements::anglo_egyptian_turn_four_is_all_remaining](#)
- [omdurman-rules::src::effects::tests::campaign_reinforcement_cap_and_double_entry](#)
- [omdurman-rules::src::effects::tests::campaign_gunboats_quota_three_per_turn](#)
- [omdurman-rules::src::effects::tests::reinforcement_rejected_onto_enemy_occupied_hex](#)
- [omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force](#)

§9.211 – Anglo-Egyptian set up first, moves second (Historical)

implemented

manual page unknown

The Anglo-Egyptian player sets up first, and moves second:

- Not in play: GORDON leader unit, “Friendlies” brigade.
- Gunboats start in any Nile River hexes adjacent to the Zariba.
- Camel Corps, Egyptian Cavalry, and Horse Artillery start in the village of Kerreri hut hexes.
- All remaining Anglo-Egyptian units set up in the 13 hexes of the Zariba.

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:346 GH:omdurman-rules/src/effects/dispatch.rs:346	first_player	<pre> 344 345 /// The player who moves first in a scenario (§4, §9.113, §9.212, §9.322). 346 pub fn first_player(scenario: Scenario) -> Player { 347 match scenario { 348 Scenario::Campaign => Player::AngloEgyptian, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::historical_setup_rejects_not_in_play_units](#)

§9.212 – Dervish set up (Historical) – deployment zones and leader-to-hex mapping

implemented

manual page unknown

The Dervish player sets up second, and moves first.

- Not in play: Isa Zachneih, gunboats, and forts.

- All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.
- Dervish leaders start on the lettered hexes:
 - A: Ali Wad Helu
 - D: Sheik El Din
 - Y: Yakub
 - K: Khalifa Abdullah
 - S: Sherif

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:468 GH:omdurman-rules/src/effects/state.rs:468	in_deployment_zone	<pre> 466 /// plan / UI rather than this hex predicate. Documented, not silently 467 /// dropped. 468 pub fn in_deployment_zone(&self, player: Player, hex: HexCoord, is_boat: bool) -> bool { 469 // No board attached -> permissive (unit tests, unbound session). 470 if self.board.terrain.is_empty() { </pre>
omdurman-rules/src/lib.rs:387 GH:omdurman-rules/src/lib.rs:387	DervishLeader::setup_letter	<pre> 385 /// (§9.212): A→Ali Wad Helu, D→Sheik El Din, Y→Yakub, K→Khalifa Abdullah, 386 /// S→Sherif, O→Osman Digna. Inverse of [DervishLeader::setup_letter]. 387 pub fn setup_letter(self) -> SetupLetter { 388 match self { 389 DervishLeader::AliWadHelu => SetupLetter::A, </pre>
omdurman-rules/src/lib.rs:403 GH:omdurman-rules/src/lib.rs:403	dervish_leader_for_setup	<pre> 401 /// inherent impl here, so the mapping is a free function -- the bijective 402 /// inverse of [DervishLeader::setup_letter]. 403 pub fn dervish_leader_for_setup_letter(letter: SetupLetter) -> DervishLeader { 404 match letter { 405 SetupLetter::A => DervishLeader::AliWadHelu, </pre>

Covered by tests:

- [omdurman-rules::src::effects::tests::deploy_rejected_outside_zone](#)
- [omdurman-rules::src::unit_profiles::embedded_leaders_resolve_from_their_host_section](#)
- [omdurman-app::src::scenario_setup::historical_places_all_six_leaders_when_anchors_present](#)
- [omdurman-app::src::scenario_setup::missing_anchor_is_reported_not_dropped_silently](#)
- [omdurman-rules::src::effects::tests::setup_letter_dervish_leader_roundtrip](#)
- [omdurman-rules::src::effects::tests::setup_letter_to_dervish_leader_known_values](#)
- [omdurman-rules::src::effects::tests::historical_setup_rejects_not_in_play_units](#)

§9.231 – Thorn hedge hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:939 GH:omdurman-rules/src/lib.rs:939	ZaribaThornHedge	<pre> 937 Terrain(i16), 938 /// -2 thorn-hedge defensive modifier (§9.231). 939 ZaribaThornHedge, 940 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 941 /// units (those Nile-side of the trench hexside). </pre>
omdurman-types/src/lib.rs:206 GH:omdurman-types/src/lib.rs:206	ZaribaThornHedge	<pre> 204 Crest, 205 /// Historical-scenario thorn-hedge segment of the Zariba (§9.231). 206 ZaribaThornHedge, 207 /// Historical-scenario trench segment of the Zariba (§9.232). 208 ZaribaTrench, </pre>

omdurman-rules/src/effects/state.rs:282 GH:omdurman-rules/src/effects/state.rs:282	has_zariba_thorn_hedge	<pre> 280 /// *effective* hexsides, so constructed (§5.3) and authored hedges both 281 /// count. 282 pub fn has_zariba_thorn_hedge(&self, hex: HexCoord) -> bool { 283 for n in hex.neighbors() { 284 if self.hexside_effective_is(hex, n, k k == HexsideKind::ZaribaThornHedge) { </pre>
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Covered by tests: [omdurman-rules::src::effects::tests::zariba_fire_penalties_apply_to_dervish_fire_only](#)

§9.232 – Trench hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:942 GH:omdurman-rules/src/lib.rs:942	ZaribaTrenchEntrenched	<pre> 940 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 941 /// units (those Nile-side of the trench hexside). 942 ZaribaTrenchEntrenched, 943 } 944 </pre>
omdurman-types/src/lib.rs:208 GH:omdurman-types/src/lib.rs:208	ZaribaTrench	<pre> 206 ZaribaThornHedge, /// Historical-scenario trench segment of the Zariba (§9.232). 208 ZaribaTrench, /// One of the two end hexsides of a Zariba trench segment that connect to 210 /// the Nile River (§9.233). Units may only enter/ leave the Zariba via </pre>

Proven by: [omdurman-rules::src::lib::melee_modifier_keeps_roll_legal](#)

Covered by tests: [omdurman-rules::src::effects::tests::trench_entrenched_units_take_trench_modifiers](#)

§9.233 – Zariba entry/exit costs

implemented

manual page unknown

Units may only enter and/or leave the Zariba via the two end hexsides that connect to the Nile River, paying +2 movement points to cross (Exception: advance after combat across an entrenched hexside).

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:258 GH:omdurman-types/src/lib.rs:258	blocks_movement	<pre> 256 /// `omdurman-rules`). The trench *end* variants are therefore intentionally 257 /// not blocking. 258 pub fn blocks_movement(self) -> bool { 259 matches!(260 self, </pre>
omdurman-rules/src/effects/state.rs:295 GH:omdurman-rules/src/effects/state.rs:295	zariba_entry_surcharge	<pre> 293 /// +2 movement points to cross"). Trench ends are authored FoK geography 294 /// (players cannot construct them), so this reads the authored board. 295 pub fn zariba_entry_surcharge(&self, from: HexCoord, to: HexCoord) -> i16 { 296 match self.board.hexside_between(from, to) { 297 Some(k) if k.is_zariba_trench_end() => 2, </pre>
omdurman-rules/src/effects/	movement_cost_for	<pre> 964 /// 965 /// §5.42: entering or leaving an enemy ZOC adds no </pre>

state.rs:966 GH:omdurman-rules/src/effects/state.rs:966		MP cost. 966 pub fn movement_cost_for(967 &self, 968 unit: &UnitPlacement,
omdurman-rules/src/effects/movement.rs:31 GH:omdurman-rules/src/effects/movement.rs:31	apply_move_unit	29 /// the true terrain cost (\$5.11) and enforces gunboat upstream/downstream 30 /// allowances (\$5.24); otherwise it falls back to the caller-supplied `cost`. 31 pub fn apply_move_unit(32 state: &mut GameState, 33 unit_id: UnitId,

Proven by: omdurman-types::src::lib::zariba_trench_ends_differ_only_in_the_entry_rule

Covered by tests: omdurman-rules::src::effects::tests::zariba_end_hexside_costs_extra_mp
omdurman-rules::src::effects::tests::zariba_thorn_hedge_blocks_movement

§9.321 – British set up (Bonus)

implemented

manual page unknown

The British player sets up first, moves second:

- General GORDON leader unit in the palace.
- Two old style (unnamed) gunboats in any Nile River hexes.
- Set up in any building or hut hexes of Khartoum, Forts Makran and/or Buri, and/or adjacent to any wall hex:
 - one Egyptian Battalion artillery unit
 - two British infantry units (represents Caucasian troops)
 - three Egyptian infantry units (represents Cairo “Bazouks”)
 - four Sudan infantry units (represents Sudanese blacks)
 - four “Friendlies” units (represents the Shaggyeh)

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/scenario_setup.rs:42 GH:omdurman-rules/src/scenario_setup.rs:42	FALL_OF_KHARTOUM	40 /// North Fort uses a campaign HadendowaForts counter (one of the spare fort 41 /// sprites). 42 pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[43 FixedPlacement { 44 section: SectionName::BritishBoats,

Covered by tests: omdurman-rules::src::effects::tests::confirm_ready_rejected_below_scenario_target
omdurman-rules::src::effects::tests::remove_deployed_unit_happy_path
omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace
omdurman-app::src::scenario_setup::fall_of_khartoum_reports_missing_palace
omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes
omdurman-rules::src::effects::tests::fok_ae_gunboat_deploys_only_on_nile
omdurman-rules::src::effects::tests::fok_ae_land_unit_rejected_on_nile
omdurman-rules::src::effects::tests::fok_setup_complete_requires_full_oob
omdurman-rules::src::effects::tests::deploy_via_real_sprite_resolution_matches_engine
omdurman-rules::src::effects::tests::british_boats_named_vs_old_gunboat_detection
omdurman-rules::src::effects::tests::fok_order_of_battle_british omdurman-rules::src::effects::tests::fok_order_of_battle_dervish
omdurman-rules::src::effects::tests::fok_caps_bind_across_counter_variants

§9.322 – Dervish enters turn one (Bonus)

implemented

manual page unknown

Dervish player moves first: enters turn one through any hexes on the south or east edge of the map.

- 32 Mulazmin units (represents combined forces of Wad El Nejumi, Abu Girgeh, and Sheik El Obeid)
- 2 Hadendowa; 6 Kehena; 5 Degheim (represents Mahdi's combined west bank forces)
- 3 Dervish artillery units.

File	Symbol	Code Snippet
omdurman-rules/src/unit_profiles.rs:392 GH:omdurman-rules/src/unit_profiles.rs:392	ali_wad_helu	<pre>390 /// §9.2/§9.3) unplaceable -- the FoK order-of-battle table 391 /// (`fok_cap_group`) is keyed by identity. 392 pub fn ali_wad_helu(col: u32, row: u32) -> Option<Classification> { 393 match (col, row) { 394 (0, 0) => dervish_leader(DervishLeader::AliWadHelu),</pre>
omdurman-types/src/lib.rs:846 GH:omdurman-types/src/lib.rs:846	sections_for_picker	<pre>844 /// counter of those two forces and leave the §9.322 entry force -- and 845 /// with it the setup Ready gate (§9.2/§9.3) -- unplaceable. 846 pub fn sections_for_picker(self) -> Option<&'static [SectionName]> { 847 match self { 848 Scenario::Campaign Scenario::Historical => None,</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::ali_wad_helu_block_resolves_leader_and_degelim_tribes](#)
[omdurman-rules::tests::fok_setup_flow::degheim_counters_resolve_to_the_degheim_tribe](#)
[omdurman-rules::src::effects::tests::fok_setup_complete_requires_full_oob](#)
[omdurman-rules::src::effects::tests::fok_dervish_land_unit_rejected_on_nile](#)
[omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks](#)
[omdurman-rules::src::effects::tests::fok_order_of_battle_dervish](#)
[omdurman-rules::src::effects::tests::fok_dervish_east_edge_on_diamond_board](#)

§9.341 – Turn 1 is always a night turn (Bonus)

implemented

manual page unknown

Turn 1 is always a night turn (see 8.1).

See also: §8.1

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:172 GH:omdurman-rules/src/turn_track.rs:172	FALL_OF_KHARTOUM	<pre>170 /// (the rulebook fixes none); only `day_night` is rule-bearing (night halves 171 /// Anglo-Egyptian movement and ranges and bars howitzer fire, §8.1). 172 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [173 TurnEntry { 174 turn: 1,</pre>

Proven by: [omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario](#)
[omdurman-rules::src::turn_track::fall_of_khartoum_night_window_is_turns_one_and_two](#)

Covered by tests: [omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night](#)

§9.342 – All hexes are playable, including half hexes (Bonus)

implemented

manual page unknown

All hexes are playable, including hexes showing up half or less.

File	Symbol	Code Snippet
omdurman-rules/src/board_data.rs:37 GH:omdurman-rules/src/board_data.rs:37	fall_of_khartoum_map	<pre>35 36 /// The Fall-of-Khartoum board (§9.3). 37 pub fn fall_of_khartoum_map_data() -> MapData { 38 FOK.get_or_init(parse(FOK_RON)).clone() 39 }</pre>

Covered by tests: [omdurman-rules::src::effects::tests::fall_of_khartoum_board_excludes_no_hexes](#)

§9.343 – Both players use the Dervish Range Effects Table (Bonus)

implemented

manual page unknown

Both players must use the Dervish Range Effects Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects/fire.rs:53 GH:omdurman-rules/src/effects/fire.rs:53	range_band_for	<pre>51 /// in FALL OF KHARTOUM *both* players use the Dervish Range Effects Table 52 /// (§9.343). 53 pub fn range_band_for(54 scenario: Scenario, 55 player: Player,</pre>

Proven by: [omdurman-rules::src::effects::fire::range_band_for_routes_to_the_right_faction_table](#)
[omdurman-rules::src::effects::fire::range_table_player_for_routes_friendlyes_and_fok_to_the_dervish_table](#)

Covered by tests: [omdurman-rules::src::effects::tests::fok_both_players_use_dervish_range_table](#)

§9.344 – Dervish controls the North Fort (Bonus)

implemented

manual page unknown

The Dervish player controls the “North Fort” and may fire its guns.

File	Symbol	Code Snippet
omdurman-rules/src/scenario_setup.rs:42 GH:omdurman-rules/src/scenario_setup.rs:42	FALL_OF_KHARTOUM	<pre>40 /// North Fort uses a campaign HadendowaForts counter (one of the spare fort 41 /// sprites). 42 pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[43 FixedPlacement { 44 section: SectionName::BritishBoats,</pre>
omdurman-rules/src/effects/state.rs:1505 GH:omdurman-rules/src/effects/state.rs:1505	hex_has_enemy_fort	<pre>1503 /// may neither occupy an enemy fort nor advance after combat into one 1504 /// (forts are never captured -- only destroyed, §6.62/§6.53/§7.6). 1505 pub fn hex_has_enemy_fort(&self, hex: HexCoord, mover: Player) -> bool { 1506 self.units.iter().any(u { 1507 u.position == hex</pre>

Covered by tests: [omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes](#)
[omdurman-app::src::scenario_setup::placement_done_gate_matches_by_identity_not_allocated_id](#)

omdurman-rules::src::effects::tests::fok_order_of_battle_dervish
omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks

§9.345 – Gunboat White Nile <-> Blue Nile crossing (Bonus)

implemented

manual page unknown

The British gunboats may move from the White Nile to the Blue Nile and vice-versa at an off-board movement cost of six “upstream” movement points.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:1489 GH:omdurman-rules/src/effects/state.rs:1489	is_nile_mouth_crossing	<pre>1487 /// must be named on the board, else this is 1488 `false` and the move falls 1489 /// through to the ordinary contiguous-Nile rules. 1489 pub fn is_nile_mouth_crossing(&self, from: HexCoord, to: HexCoord) -> bool { 1490 let white = self 1491 .board</pre>
omdurman-types/src/lib.rs:540 GH:omdurman-types/src/lib.rs:540	Location::WhiteNileMouth	<pre>538 /// The off-board mouth of the White Nile branch (FALL OF KHARTOUM §9.345) -- 539 /// a British gunboat may cross to the Blue Nile mouth for 6 upstream MP. 540 WhiteNileMouth, 541 /// The off-board mouth of the Blue Nile branch (FALL OF KHARTOUM §9.345). 542 BlueNileMouth,</pre>

Covered by tests: omdurman-rules::src::effects::tests::fok_gunboat_crosses_between_nile_mouths

§9.346 – GORDON immobile, eliminated only at the Palace (Bonus)

implemented

manual page unknown

The GORDON leader unit starts in the palace and may not move during the scenario. He may only be eliminated by a Dervish unit passing through or occupying the palace hex (as normal movement or as advance after combat).

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:482 GH:omdurman-rules/src/effects/dispatch.rs:482	check_gordon_palace	<pre>480 /// after combat). Records the turn (which fixes the §9.35 victory level) and 481 /// ends the game. A no-op outside FoK, or once GORDON is already gone. 482 pub fn check_gordon_palace(state: &mut GameState) { 483 if state.scenario != Scenario::FallOfKhartoum state.gordon_eliminated_turn.is_some() { 484 return;</pre>
omdurman-rules/src/lib.rs:644 GH:omdurman-rules/src/lib.rs:644	UnitIdentity::is_gordon	<pre>642 /// Whether this is the GORDON leader unit (§9.32, §9.346) -- the immobile 643 /// palace defender whose elimination ends FALL OF KHARTOUM (§9.35). 644 pub fn is_gordon(&self) -> bool { 645 matches!(646 self,</pre>

Covered by tests: omdurman-rules::src::unit_profiles::gordon_is_an_immobile_british_leader

omdurman-rules::src::effects::tests::gordon_survives_means_no_elimination

omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace

7/10 implemented

§10 – Optional Rules

§10 – Optional Rules

descriptive

manual page unknown

Optional Rules (Campaign game only)

It is suggested that the most intriguing employment of the following two options is to permit the Dervish player to have either one or the other, but the Anglo-Egyptian player doesn't know which one until he stumbles onto it. Players are advised that the employment of both optionals in the same game is not recommended.

§10.1 – River Mines

descriptive

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River Mines

The Khalifa twice tried (unsuccessfully) to submerge a powerful mine in the Nile to sink or damage British gunboats. This option assumes that both attempts were successful.

§10.2 – River Chain

descriptive

manual page unknown

River Chain

The Khalifa also tried (also unsuccessfully) to string a heavy chain across the Nile to stop or slow down the British gunboats. This option assumes the chain was emplaced.

§10.11 – Secretly record mine hexes

implemented

manual page unknown

Prior to the commencement of play the Dervish player secretly records two Nile River hexes to be mined (the mines may not both be placed in the same hex). These hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects/setup.rs:326 GH:omdurman-rules/src/effects/setup.rs:326	apply_place_mine	<pre>324 /// Lay a river mine during setup (§10.11). Validated by 325 /// [<code>GameState::can_place_mine`</code>]. 326 pub fn apply_place_mine(state: &mut GameState, hex: HexCoord) -> Result<(), RuleError> { 327 state.can_place_mine(hex)?; 328 state.mines.push(MinePlacement {</pre>
omdurman-rules/src/effects/state.rs:189 GH:omdurman-rules/src/effects/state.rs:189	GameState::mines	<pre>187 friendlies_transport: None, 188 optional_rules: Vec::new(), 189 mines: Vec::new(), 190 chain: None, 191 board: Arc::new(BoardInfo::default()),</pre>
omdurman-rules/src/	OptionalRule	<pre>314 /// two should be in play (rulebook §10). 315 #[derive(Serialize, Deserialize, Clone, Copy,</pre>

lib.rs:316 GH:omdurman-rules/src/lib.rs:316		PartialEq, Eq, Debug)) 316 pub enum OptionalRule { 317 RiverMines, 318 RiverChain,
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Covered by tests: [omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup](#)
[omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule](#)

§10.12 – Mine resolution

implemented

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When a British gunboat enters a mined hex, the Dervish player must order it to stop as it has struck a mine. The Dervish player then resolves the effect of the mine's blast by rolling the ten-sided die:

- 1–4: No effect
- 5–7: Gunboat damaged, lost use of its engines and must drift two hexes per turn (with the current) for the rest of the game. No effect on guns or Maxims unless they drift out of range.
- 8–10: Gunboat sunk!

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:200 GH:omdurman-rules/src/effects/effect.rs:200	RiverMine	198 // -- Optional rules ----- 199 /// River mine resolution (rulebook §10.12). 200 RiverMine { 201 gunboat_id: UnitId, 202 hex: HexCoord,
omdurman-rules/src/effects/river.rs:152 GH:omdurman-rules/src/effects/river.rs:152	apply_river_mine	150 151 /// Apply a river-mine resolution (rulebook §10.12). 152 pub fn apply_river_mine(153 state: &mut GameState, 154 gunboat_id: UnitId,
omdurman-rules/src/lib.rs:1120 GH:omdurman-rules/src/lib.rs:1120	MineResult	1118 /// British gunboat enters a mined hex. 1119 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1120 pub enum MineResult { 1121 /// Roll 1-4: no effect. 1122 NoEffect,

Proven by: [omdurman-rules::src::lib::mine_result_bands_match_the_printed_rule](#)
[omdurman-rules::src::effects::river::river_mine_triggers_even_on_a_harmless_roll](#)
[omdurman-rules::src::effects::river::river_mine_engine_loss_arms_the_drift_flag](#)
[omdurman-rules::src::effects::river::river_mine_sinking_removes_the_gunboat](#)

Covered by tests: [omdurman-rules::src::effects::tests::mine_fires_once_and_spare_dervish](#)

§10.13 – Mines consumed after both rolled for

implemented

manual page unknown

After both mines have been rolled for, no more are available.

File	Symbol	Code Snippet
omdurman-rules/src/effects/	apply_river_mine	150 151 /// Apply a river-mine resolution (rulebook §10.12).

river.rs:152 GH:omdurman-rules/src/effects/river.rs:152		152 <code>pub fn apply_river_mine(</code> 153 <code>state: &mut GameState,</code> 154 <code>gunboat_id: UnitId,</code>
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Proven by: [omdurman-rules::src::effects::river::river_mine_triggers_even_on_a_harmless_roll](#)
[omdurman-rules::src::effects::river::a_triggered_mine_never_fires_again](#)

Covered by tests: [omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish](#)

§10.14 – Dervish gunboats pass safely

[implemented](#)

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The Dervish player’s gunboats may pass through the mined hexes with no ill effect (he knows where the mines are).

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:152 GH:omdurman-rules/src/effects/river.rs:152	apply_river_mine	150 151 <code>/// Apply a river-mine resolution (rulebook §10.12).</code> 152 <code>pub fn apply_river_mine(</code> 153 <code>state: &mut GameState,</code> 154 <code>gunboat_id: UnitId,</code>

Proven by: [omdurman-rules::src::effects::river::dervish_gunboats_pass_mined_hexes_unharmd](#)

Covered by tests: [omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish](#)

§10.21 – Secretly record chain hexes

[implemented](#)

manual page unknown

Prior to the commencement of play the Dervish player secretly records a line of river hexes (not exceeding four hexes long) across which the chain is strung. The hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:732 GH:omdurman-rules/src/effects/state.rs:732	can_place_chain	730 <code>/// Read-only check of a river-chain placement in</code> 731 <code>setup (§10.21): Setup phase</code> 732 <code>/// and at most [<code>MAX_CHAIN_HEXES</code>] hexes.</code> 733 <code>pub fn can_place_chain(&self, hexes: &[HexCoord]) -</code> 734 <code>> Result<(), RuleError> {</code> 735 <code>self.require_setup_phase()?;</code> 736 <code>// Optional-rule gate: the chain exists only</code> 737 <code>when the River Chain option</code>
omdurman-rules/src/effects/setup.rs:337 GH:omdurman-rules/src/effects/setup.rs:337	apply_place_chain	335 <code>/// Lay (or replace) the river chain during setup</code> 336 <code>(§10.21). Validated by</code> 337 <code>/// [<code>GameState::can_place_chain</code>].</code> 338 <code>pub fn apply_place_chain(state: &mut GameState, hexes:</code> 339 <code>&[HexCoord]) -> Result<(), RuleError> {</code> 340 <code>state.can_place_chain(hexes)?;</code> 341 <code>state.chain = Some(ChainPlacement {</code>
omdurman-rules/src/effects/state.rs:1963 GH:omdurman-rules/src/effects/state.rs:1963	MAX_CHAIN_HEXES	1961 1962 <code>/// Maximum contiguous Nile hexes the river chain may</code> 1963 <code>span (§10.21).</code> 1964 <code>pub const MAX_CHAIN_HEXES: usize = 4;</code>

		1965 //
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Covered by tests: `omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup`
`omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule`

§10.22 – Gunboat stops on chained hex

implemented

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When a British gunboat enters a “chained” river hex it must stop and may move no further that turn.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:1005 GH:omdurman-rules/src/effects/state.rs:1005	<code>can_move_gunboat</code>	<pre>1003 /// upstream movement allowance is their maximum for that turn." Chained Nile 1004 /// hexes stop the gunboat (§10.22). 1005 pub fn can_move_gunboat(1006 &self, 1007 unit_id: UnitId,</pre>

Covered by tests: `omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk`

§10.23 – Sinking the chain

implemented

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No gunboats (British or Dervish) may cross the chain until it has been sunk by the British player. He may sink the chain by a) having an infantry or cavalry unit spend one complete turn on either riverbank adjacent to a “chained” river hex, or b) firing at the chain with artillery and achieving a 3 or more on the Combat Results Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:204 GH:omdurman-rules/src/effects/river.rs:204	<code>apply_sink_chain</code>	<pre>202 /// Sink the river chain (rulebook §10.23). Marks the placed chain cleared so it 203 /// no longer stops gunboats (§10.22). 204 pub fn apply_sink_chain(state: &mut GameState) -> Result<(), RuleError> { 205 match state.chain.as_mut() { 206 Some(chain) if !chain.sunk => {</pre>

Covered by tests: `omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk`

0/1 implemented

§11 – Historical Notes

§11 – Historical Notes

descriptive

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Historical Notes

In 1881 Mohammed Ahmed Ibn Al-Sayid Abdullah, the son of an obscure carpenter in the hinterlands of the Sudan, proclaimed himself the “Mahdi” — the Messiah of the Islamic faith. His timing was propitious indeed. Since the early 1820’s a corrupt Egypt, with the Sultan of Turkey’s blessing, had incessantly raped the Sudan, taking ivory and flooding the slave markets with some half million captured Sudanese blacks. By 1880, nearly 40,000 Egyptian troops occupied outposts scattered throughout the Sudan, enforcing Egypt’s hold on this lucrative ivory and slave trade and squeezing the native population dry through vicious and corrupt tax officials. All was controlled from Khartoum via the office of Governor General of the Sudan. The title had been held by a succession of individuals, including General Charles Gordon, whose appointment was an attempt to reinstate some rudimentary justice in the Sudan after France and Britain assumed joint political control of a bankrupt Egypt.

By 1881, however, Gordon’s term had expired and a new Governor General, again corrupt and incompetent,
(see manual for full text)

See also: §9.1, §9.2, §9.3

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Credits

§Credits – Credits

[descriptive](#)

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■ 1/1 implemented

Combat Results Table (shared reference)

§CRT – Combat Results Table (shared by §6.22 fire and §7.7 melee)

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File	Symbol	Code Snippet
omdurman-rules/src/combat_results_table.rs:10 GH:omdurman-rules/src/combat_results_table.rs:10	FireFactorRow	<pre>8 serde::Serialize, serde::Deserialize, Clone, Copy, 9 PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 10 pub enum FireFactorRow { 11 /// 1-5 factors 12 Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:46 GH:omdurman-rules/src/combat_results_table.rs:46	from_total	<pre>44 45 /// Determine which row a given total fire factor 46 pub fn from_total(total: u16) -> Self { 47 match total { 48 0..=5 => FireFactorRow::Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:88 GH:omdurman-rules/src/combat_results_table.rs:88	combat_results_table	<pre>86 /// D = `Disrupt` (1/2 of target units, round up) 87 /// 1...5 = `Eliminate(n)` (that many units removed) 88 pub fn combat_results_table(row: FireFactorRow, roll: 89 crate::tables_data::CRT[row.index()](roll.value() 90) as usize] {</pre>
omdurman-rules/src/lib.rs:1008 GH:omdurman-rules/src/lib.rs:1008	CombatResult	<pre>1006 /// * `--` -- no effect 1007 #[derive(Serialize, Deserialize, Clone, Copy, 1008 PartialEq, Eq, Debug)] 1009 pub enum CombatResult { 1010 NoEffect, 1011 Disrupt,</pre>

Proven by: [omdurman-rules::src::lib::disrupt_half_is_rounded_up](#)
[omdurman-rules::src::combat_results_table::crt_eliminate_count_stays_within_printed_bounds](#)
[omdurman-rules::src::combat_results_table::crt_is_monotone_in_the_die_roll_for_every_row](#)

Covered by tests: [omdurman-rules::src::combat_results_table::ae_combat_results_table_lowest_is_no_effect](#)
[omdurman-rules::src::combat_results_table::ae_combat_results_table_highest_is_eliminate_5](#)
[omdurman-rules::src::combat_results_table::ae_combat_results_table_progresses_with_roll](#)
[omdurman-rules::src::combat_results_table::ae_combat_results_table_progresses_with_factor](#)
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[omdurman-rules::src::combat_results_table::crt_all_rows_monotone_non_decreasing](#)
[omdurman-rules::src::combat_results_table::crt_cross_row_monotone_for_each_roll](#)
[omdurman-rules::src::combat_results_table::crt_lowest_row_is_worst_highest_row_is_best](#)
[omdurman-rules::src::combat_results_table::crt_eliminate_never_exceeds_5](#)
[omdurman-rules::src::combat_results_table::crt_every_cell_matches_the_table](#)

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| Reference – Charts and Tables

§Reference – Charts and Tables

out-of-scope

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